

This is an excerpt from Frank Elavsky's dissertation on *Tool-making as an Intervention on the Accessibility of Interactive Data Experiences*, which can be accessed in full at this archival link (once published):

<http://reports-archive.adm.cs.cmu.edu/anon/hcii/CMU-HCII-26-103.pdf>

This document contains the following sections:

- Abstract
- Chapter 1: Motivation (Section 1.1 only)
- Chapter 2: Background & Related Work
- Chapter 3: Overview of Contributions
- Chapter 9: Findings (Sections 9.1, 9.2, and 9.6 only)
- Chapter 10: Discussion & Future Work (Sections 10.2 and 10.3 only)
- References

This document does not contain the following chapters:

- Chapter 4: *Chartability*: Heuristics as a Tool and Resource
- Chapter 5: *Data Navigator*: Low-level Tooling for Creating Navigable Data Structures
- Chapter 6: *Skeleton*: Visual Authoring of Non-visual Data Experiences
- Chapter 7: *Cross-perception*: Rethinking Input Design Towards Blind Analytical Interaction
- Chapter 8: *Softerware*: Enabling Personalization of Interactive Data Representations for Users with Disabilities
- Chapter 12: The *Softerware* Option Taxonomy
- Chapter 11: Biographical Sketch

Abstract

In this dissertation, we contribute practical advancements in tool-making as an intervention on the accessibility of interactive data experiences. The thesis of this dissertation is as follows:

This dissertation argues that the tools practitioners use to build interactive data experiences are themselves sites where accessibility barriers are produced, prevented, or alleviated for both end users and authors. This work contributes five tools, Chartability, Data Navigator, Skeleton, Cross-perception, and Softeware, that collectively center accessibility work on the empowerment of disabled and non-disabled practitioners across the full arc of evaluation, data navigation, analytical interaction, and personalization.

Rather than framing accessibility research solely around ideal experiences for end users with disabilities, this thesis investigates why accessibility work is so difficult for the practitioners who build interactive data experiences and what tool-making can reveal about those difficulties. We organize this investigation around four domains where practitioners face the most persistent challenges: evaluation, navigation, interaction, and personalization. *Chartability*, a heuristic framework, contributed first and mapped the full landscape of accessibility barriers. *Chartability* helped us to identify and prioritize our three latter domains (navigation, interaction, and personalization) as the areas where the most severe gaps remain in practitioner work. *Data Navigator* and *Skeleton* then investigate navigation, finding that practitioners struggle because navigation structure has no visible, manipulable representation in their workflows. *Cross-perception* engages interaction, demonstrating that blind data analysis has been constrained by existing tools and that a new interaction design framework can reshape what analytical work is possible. *Softeware* addresses personalization, revealing that access needs genuinely conflict across users and that meaningful personalization requires system-level infrastructure that does not yet exist in practitioner tooling ecosystems.

Combined, these contributions provide empirical insights and practical advancements in the state of the art for tooling that bridges gaps in current accessibility practices in visualization and data science. Our work ultimately enables people with and without disabilities to better evaluate barriers in, analyze with, design for, develop, and personalize interactive data experiences. We demonstrate that tool-making is a productive intervention that both engages accessibility barriers and elucidates why those gaps exist in practitioner work.

Part I

Introduction

Chapter 1

Motivation

Technology grows, develops, and adapts alongside humanity. Sometimes, we adapt or appropriate existing technologies to our needs. And in the last two decades, “data” (as a social and cultural vehicle for transferring information through computers) has become a way for everyday people to gain a greater understanding of the world they participate in, even *deepening* their participation.

Data is used to help people make better personal financial decisions, to learn about their health, to learn about the health of societies, to pick who to vote for, to understand trends, wars, and nearly every major issue of daily life. The magnitude to which data now participate in, or are fundamental as a substrate, for a person’s health, happiness, and agency cannot be understated.

Data, and access to it, is paramount. Knowing how to read data, read visualizations, interact with data systems, manipulate data interfaces; this unlocks many opportunities for people, such as gaining job opportunities, having more confidence, and communicating more effectively.

This thesis aims to address a broader injustice that has arisen alongside data as an emerging source of agency in people’s everyday lives: that some people, namely people with disabilities, often face barriers to accessing data, data interfaces, and data visualizations in particular.

People with a broad range of visual disabilities face the most pervasive forms of exclusion when interacting with data, because *visualizations* are the most common form of data representation. Visual disabilities include people who are “blind” (which may be congenital, progressive, total, partial, “legally” as a designation, and so on; a whole spectrum), people who are “low vision” (which may include forms of night blindness, light sensitivity, visual obstruction, limited field of view, general lack of acuity, color vision deficiency, and so on; another whole spectrum), as well as those who might have visual-cognitive disabilities, such as epilepsy, vestibular vision disorders, and more. All of these people may experience partial or total exclusion from data visualizations and interactive data interfaces and experiences. Some may have access *technically speaking*, but the access is arduous, painful, and nearly unusable in practice.

And, as we unpack later in Chapter 4, many other people with disabilities other than those related to vision also experience barriers when interacting with data.

To no fault of these people, they are not given equal, equitable, or just agency to use data in their lives. We refer to this broader phenomenon as “inaccessibility.”

“Inaccessibility” is when a person or system creates something that contains a barrier that excludes a person with a disability. And if a clever reader carefully inspects the previous sentence, we can begin to understand who is first responsible for making something accessible (hint: *not* the person who has been excluded!). Upon recognizing that the creators are first responsible, this then helps us identify which new challenges and problems arise that we must address in order to make accessibility a reality.

In the following section of this thesis’ motivation, we add more clarification (as a necessary pre-requisite) for better understanding not just what we are doing in this thesis, but how we frame our work and situate the “researcher” as an actor who has power to enact change towards more accessible data experiences for people with disabilities.

1.1 What do we mean by “accessibility” and “tools?”

This thesis is a body of research situated within the existing research area focused on making interactive data representations more accessible for people with disabilities. Much of the work in this existing area is situated within the context of making interactive data *visualizations* accessible, particularly (but not exclusively) for people who are blind. Our work, contributed here in this thesis, is focused on using *tools* as a specific intervention and sub-area of study for making interactive data *representations* accessible for people with disabilities, broadly speaking. (“Representations” here is an intentionally broader term than “visualizations,” which are exclusively visual representations of data.)

Before we begin, two things must be understood up front, or else the rest of this thesis could be interpreted with disruptive assumptions: we must interrogate the phrase “making visualizations accessible” and unpack why *tools* are a meaningful area of study.

1.1.1 Is “accessible visualization” really an oxymoron?

The first assumption that must be disrupted is perhaps the motivating cornerstone of this research, which is that the phrase “making visualizations accessible,” while a noble goal, is not the semantically correct phrasing nor precisely what describes our work. This can be misleading. We do use the phrase “accessible visualization” but will admit that this seems to confuse certain people with very particular opinions about things. We will clear this up.

Villains of our field’s past have written incendiary and ableist perspectives on why “no forms of data visualization, not just dashboards jam-packed with graphics, can be made fully accessible to someone who is blind,” and that “[a blind man] will never be able to analyze data as I do visually, because many aspects of vision cannot be duplicated by his other senses” [30]. However, this position misunderstands what the goal of accessibility is, and arguably even what the goal of visualization itself is.

Making visualizations accessible *isn’t* about the visualization, it’s about making the outcomes of the visualization accessible.

Visualizations are ubiquitous and paramount for decision-making. However, the *artifact* that is a visualization is not even the goal of the act of visualizing: developing understanding, insight, confidence, and communication among and between human beings are the goals of visualization. Visualization is about making data easier to use for all kinds of things. Yes, our visual system enables us more than any other form of sensory cognition that we have [18, 37, 118]. But we aren’t trying to make sight itself accessible. We are trying to make it possible for people to make meaningful decisions, gain valuable information, build conjectures, and effectively communicate with others.

Many, many people who I, Frank, have spoken to over the course of my career, even before embarking on this thesis journey, misunderstand this simple fact: making a “visualization” accessible *isn’t* about the visualization itself but rather making what the visualization is meant to *accomplish* accessible. It’s about equal outcomes, not equal interactions with an artifact.

People with disabilities are no small portion of the world’s population. In the United States, 26% of people self-report living with at least one disability that affects their daily lives [95] and all of us will eventually age into disability (if we are lucky to live a long life).

People with disabilities deserve to participate fully in life. They deserve financial independence. They also deserve loving care and interdependence [8, 60, 91]. People with disabilities have a right to make informed decisions, to know about the status of a global pandemic, and to have an understanding of local and national politics [29]. While we use visualizations to navigate all of these domains, the goal is not to make the charts and graphs themselves somehow equally useful to all people. That would be a false measurement of success.

Our goal then, is measured by the success of lives led by people with disabilities [123]. Many other measurements are just metrics along the journey towards that goal. We then ask: Can people with disabilities also use data to live full lives? Can they make *fast* decisions based on data? Meaningful, careful, *slow* decisions? Communicate complex ideas? Crunch and clean data, develop models, find errors, and build hypotheses? Can they have memorable, immersive, beautiful, aesthetic experiences with data too [60]? Ultimately, our aim is not accessible “visualizations” but equitable and accessible *outcomes*, everything an interactive data experience *accomplishes* for the people who use it.

Again, if the goal of accessible visualization were about visualizations themselves, then the correct course of action would be one framed by the medical model of disability [85]: that there is a normative state of behavior and capability (in this case, it would be “normal” to be able to read a visualization and make a decision) and any deviation from that norm must be corrected. This framing first assumes that the visualization should not be altered or improved. And then this framing puts the burden on the bodies of people with disabilities: that they must be “fixed” and given sight or brought to some equivalent state as someone who is “healthy,” normal, and sighted. Plenty of scholars have already discussed why this framing is a problem, not only because it places undue burden on people with disabilities and produces pathologies and hierarchies of disability, but also because it is fundamentally not economically or ethically feasible.

So we then turn to other models of disability, such as the social model. The social model is heavily discussed by disability scholars and is not the end-game or last and total way of thinking about disability [85, 98, 99, 116, 142]. But the core motivation is that society, not medicine, is also a path towards solving problems that people with disabilities face. A few important concepts and concretely actionable things come from the social model that can help motivate the work of this thesis.

First, we look to the historical birth of the social model of disability: in the 504 sit-ins that took place in the United States in 1977. Cities had curbs and curbs are a barrier for people who use wheelchairs. So protests happened because decisions were being made without people with disabilities at the table. In this instance, people acknowledged that political power was an exclusive club and fought to ensure their cry “nothing about us, without us!” materialized.

And this leads us to the first and most-foundational philosophical framing for this thesis: that our *artifacts*, these things we’ve created from curbs to data visualizations, can become *barriers* for people with disabilities. And it is then the artifact, not the body of the person with a disability, where disability is produced in this model. Rather than a comparison to a normative state as a way to frame disability (the medical model), we instead must observe and evaluate material outcomes based on human-made problems.

So, the social model is framed around society “solving” inequities: we get involved and make political and legal change tangible. But a second model also emerges from within the

social model: one where we can now frame *who is first responsible* for repair: the curb designers and implementers.

And knowing who is first responsible for access leads us into the moral and ethical imperative that motivates this thesis: the builders and makers of visualizations are ultimately the ones who provide exclusive value for only a subset of people: those *without* disabilities. **We must first change how builders and makers do their work.**

So the phrase “accessible visualization” is really about recognizing that visualizations produce barriers for people. That means that it is our ethical imperative, as builders and makers, to fix them. And that act of fixing barriers leads us away from mere visual representations of data into a wide variety of other senses and interaction modalities. There are many paths forward towards fuller and more-equitable lives led by people with disabilities.

1.1.2 On *tools*, *tool-making*, and *human-tool* interaction

Then the act of making becomes immensely important: we, the builders and makers of our world, need to get things right; there is a risk involved when making things that we will exclude people with disabilities. We need to make sure that we build a better world than the one we have now. We must care for new things we create and tend to the repair and maintenance of what we’ve already made. And this ethical imperative leads us to the topic of *tools* and *tool-making*.

So the second thing that must be understood before we embark on this thesis is that *tools* are not the same as *solutions* or *applications*. Sometimes tools can be used to *solve* things and are certainly, in ideal circumstances, *applied* in various contexts. But understanding the role of the “tool” in human-tool interaction is paramount for engaging in the work of making anything accessible for people with disabilities.

We use tools to shape our world, break old things, and make new things. But a tool, like the hammer (as an example), does not inherently *solve* something like homelessness. But a hammer can be used to build homes if there are social policies in place and proper resources allocated. This means that for the success of tooling, there is often a larger material, social, legal, and policy reality that supports and necessitates those tools. This thesis will not be focusing on changing the upstream dependencies, but optimistically operating as if they were true (or will be true in time).

However, in some cases, tools can *destroy*. The hammer has a claw and can easily pry apart boards and tear down homes. So tools carry potential to do all kinds of things, both good and bad, and how a tool is used is often open-ended, variable, and heavily dependent on socio-technical realities. Tools participate in personal and political agendas [138] and are sometimes, for this reason, regulated or made proprietary and controlled by powerful entities [40, 133].

So tools are not without any sort of ethics. We cannot just blame tool-users for outcomes when much of a tool depends on these larger systems and structures. Technologies (tools included) encode the assumptions and biases of their *creators* as much as, if not more than, their users. Tools that build things for others to use can be loaded with assumptions about what people are *able* to do [139] and also rules and guardrails about what anyone downstream from that tool’s design *should* do [40, 132]. These assumptions, biases, and rules *limit*, *enforce*, *magnify*, *exclude*, and *enable* what a tool-user is capable of.

Tools for visualizing data are a perfect case study in this problem: virtually every major data visualization library, application, or software ever made was made entirely with the assumption that data should be transformed into visual representations. This is a reasonable assumption, since virtually all of the tool-makers are sighted and visualization is incredibly helpful to our cognition of and communication with data [33].

So data visualization, as a field, has focused its tool-making efforts on reducing the difficulty involved in visualizing data. Some visualization tools are concise [105], others are lower level but much more expressive [15]. Tool-making in visualization has focused on making it easier to scaffold a wide variety of interactions both with the visualizations as well as with their underlying data models [52].

However, as time has moved on, people began to speak out about color-vision deficiency in data visualization. Some people, primarily those with X/Y chromosomes (largely men) who are of European ancestry, have a deficiency in their ability to perceive certain colors. Then a plethora of research arose that began to look into the barriers that folks who are colorblind face in data visualization. As a result, our practices and tools improved. We began to educate practitioners, develop new color palettes, researched new methods for testing our designs, and built new systems for handling automatic color encoding. Our tools evolved.

But now data visualizations have arguably become ubiquitous in daily life. By comparison, we have far more tools now for making visualizations quickly and easily than we do for representing data in non-visual ways. We also have far more research, relatively speaking, into how sighted end users interact with visualizations.

So this thesis engages gaps that arise in this space: Practitioners face immense challenges when crafting accessible data experiences. We first need to educate practitioners on what accessibility barriers actually are in interactive visualizations. Then, we must help them engage the hardest barriers in this work and create building blocks that help them to construct navigable data experiences, build design frameworks that can inform entirely new kinds of data interaction, and develop software systems for end-user personalization and agency. Our research seeks to advance the state of the art in tools that assist in accessible data interaction while also using tool-making as an intervention that helps us to better understand and characterize *why* and *how* data practitioners face barriers themselves in this work.

Chapter 2

Background & Related Work

This chapter sets up the story that the rest of this thesis engages. It makes three moves. First, we ground the idea that tools are not neutral instruments: tools shape the work and the workers who use them, which is why tool-making is a meaningful site for accessibility intervention. Second, we survey the long conversation between accessibility and data visualization and its history across modalities, including engaging critical voices that intersect with visualization and accessibility work which inform, support, and hold in tension the work of this thesis. Lastly, we argue that the hardest remaining challenges cluster in three domains, navigation, interaction, and personalization, and we signpost which chapters engage each of these.

2.1 Tools and the People Who Use Them

2.1.1 Studying Practitioners and Their Work

Human-computer interaction has a long tradition of studying the people who make things. Ethnographic and case-study methods immerse researchers in maker, DIY, and assistive technology communities to understand how practitioners actually work, especially when the work is new and unfamiliar to them [59, 61, 62]. Participatory design and co-creation bring practitioners and end users directly into the research process, surfacing how designers shift their thinking when they encounter novel problems [45, 103, 117]. And a growing body of design-based research aims not merely to fill gaps in existing practice but to extend what practitioners are capable of: scaffolding experimentation, supporting learning, and amplifying creative and technical expression [22, 43, 65, 70, 125].

This thesis sits inside that tradition but takes a particular methodological stance: we study practitioners *through their tools*. Each project in this thesis builds a tool, places it in front of working practitioners, and observes what the tool reveals about why accessibility work is hard for them. Sometimes the tool is a resource practitioners apply in their real environments, sometimes a system they build with, and sometimes a probe whose main purpose is to make an invisible part of their work observable. Tool-making here is both an intervention and an instrument of inquiry. That stance only makes sense, however, if tools really do shape work in a deep way. So before surveying the accessibility and visualization literature, we want to engage that claim seriously.

2.1.2 How Tools Use Us

We tend to talk about using tools. We talk much less about how tools use us. A chair is a tool for sitting, but a lifetime of chairs trains a body into the chair’s posture. A pickaxe is a tool for extracting ore, but the mine organizes the miner: their hours, their movements, their risks, and what counts as a day’s work. Tools come with a direction already built in, and the people who take them up inherit that direction. As a participant in Hohman et al.’s work on model compression in practice remarked, “Better tools make it easy to do correct things and harder

to do incorrect things” [58]. Tools are made precisely with the intention to shape us and our behavior.

Sara Ahmed gives this intuition theoretical teeth. In her phenomenology of orientation, objects are not passive things waiting to be used: they direct bodies, making some actions feel near and natural while other actions fall out of reach, and the social world is full of what she calls straightening devices, arrangements that keep bodies aligned with the paths already laid down for them [1]. In her later work she traces how *use* itself is formative: things become shaped by their use, “fit” for some users and not others, and a history of use leaves a trace that directs how a thing will be used next [2]. Bowker and Star make a parallel argument at the scale of infrastructure: the classification systems and standards embedded in our information systems are largely invisible scaffolding, and every standard valorizes some point of view while silencing another [16]. Assumptions about ability are among the assumptions most commonly baked in: tools encode what their makers believe people are able to do [139].

Visualization research has begun to examine its own tools in these terms. McNutt’s survey of 57 JSON-style domain-specific languages for visualization shows that each grammar embodies contested decisions about who its user is and what that user may express, with recurring tensions between formal and colloquial specification and between competing user types [88]. His dissertation develops the larger argument: the design of visualization languages, and of the tools that surround them, directly shapes and can either enhance or constrain end-user agency, motivating systems that offer assistance while respecting the user’s intent [89]. In other words, the field’s own instruments are straightening devices. Every charting grammar that assumes a sighted reader, every renderer that emits unstructured pixels, quietly directs practitioners away from accessible outcomes, not by forbidding them but by making them feel out of reach.

Tool users are not powerless in this relationship. A rich literature on appropriation documents how people adapt tools beyond their designers’ intent, and how that emergent use can be a source of design knowledge in its own right [26, 27, 102, 122], with some work building general theory from the study of emergent and generative tool use [7, 10]. HCI’s systems community has likewise developed methods for building and studying tools deliberately: piggybacking on existing systems to study an improvement in context [41, 55], and contributing toolkits whose value is argued through demonstration, usage, technical performance, or heuristics. Ledo et al., analyzing 68 toolkit papers, found these four evaluation strategies dominate, with demonstration by far the most common [75]. That finding will matter later in this chapter: a field can produce many demonstrated toolkits while producing very little maintained infrastructure.

The takeaway from this section is the premise the whole thesis rests on: tools direct work. If the tools of data visualization currently straighten practitioners toward inaccessible outcomes, then re-orienting those tools is not a cosmetic fix. It is an intervention at the place where practice is formed. The rest of this chapter examines what the field has learned about accessible data experiences, and why that learning has so far failed to re-orient the tools.

2.2 The Conversation Between Accessibility and Data Visualization

Accessibility and data representation have been in conversation far longer than the modern visualization community has existed, and that conversation has two important threads. The first is a history of techniques: a steady accumulation of ways to carry data to people through senses and devices other than a sighted eye on a screen. The second is a critical lineage, largely led by disabled scholars and disability studies, that questions how that technical work is framed, who leads it, and who it serves. This thesis tries to take both seriously, so we survey them in turn.

2.2.1 A Short History Across Modalities

Tactile representation is the oldest thread. Tactile maps and graphics for blind readers date to at least the 1830s [44], and tactile sensory substitution for charts has been studied in the computational sciences since the early 1980s [108]. Practice in this modality is unusually mature: tactile and braille standards are robust and well-maintained [6], although their assumptions (embossed paper, fixed layouts) transfer only partially to digital, interactive contexts. Research has worked on that seam, for example by augmenting tactile graphics with audio annotations [5], and industry accessibility teams have published candid experience reports on what it takes to produce tactile graphics inside a real design workflow [24]. Recent work continues to refine the perceptual foundations of non-visual structure [90].

Sonification has a similarly long arc, and because it is a substantial and active research community in its own right, it deserves explicit treatment here. Mansur et al.’s sound graphs demonstrated in 1985 that line data mapped to pitch over time could communicate trends to blind listeners [84]. Subsequent decades built out the empirical base: cross-modal comparisons of auditory and visual graphs [32], non-visual presentation of structured information [17], techniques for scanning and comparing multiple data series by ear [87], and interactive sonification of geo-referenced data [143]. The community consolidated its methods and theory in the Sonification Handbook [54]; Walker and Nees’s theory chapter formalizes sonification as a mapping problem from data dimensions to auditory dimensions, emphasizes context cues that play the role axis ticks play in vision, and stresses that interaction, not just playback, shapes what listeners can extract [129]. Sonification research has also been a site of methodological progress, including co-design with blind and low vision collaborators [23].

What is most relevant for this thesis is the recent trajectory: sonification is increasingly integrated into multimodal, interactive systems rather than offered as a standalone substitution. VoxLens adds summaries, sonification, and voice queries to web visualizations through a JavaScript plug-in, and in a study with 21 screen reader users improved information-extraction accuracy by 122% and interaction time by 36% over conventional interaction [112]. Chart Reader, co-designed with screen reader users, combines structured navigation with sonification and author-configurable data insights [124]. Auditory maps have reached usable maturity for wayfinding and choropleth reading [12]. Notably, sonification is also one of the few accessible-visualization techniques to cross into production: Apple shipped Audio Graphs in 2021 as a VoiceOver feature with a public developer API [4]. And Umwelt re-frames the relationship en-

tirely, treating sonification, textual structure, and visualization as coequal representations derived from a shared abstract model in an accessible authoring environment [146]. Sonification’s trajectory, from output substitution toward interactive structure and authoring, previews the argument this chapter is building.

Textual description is the most widely deployed modality, and research has moved it well past “write some alt text.” Lundgard and Satyanarayan’s model of semantic content distinguishes the levels of meaning a description can carry, and shows that blind and sighted readers value different levels [77]. Jung et al. offer evidence-based guidance on description length, plain language, and the ordering of information [68]. Kim et al. collected the questions screen reader users actually ask of visualizations, pointing toward more dialogic, query-driven access [72], and Chundury et al. deepen our understanding of how sensory substitution choices should be grounded in users’ practices [21].

Navigation and screen reader interaction is the youngest thread and the one closest to this thesis. Foundational work made statistical and mathematical content traversable by screen reader [38, 115]. Studies of screen reader users’ experiences with two-dimensional digital content, including visualizations, documented both the depth of exclusion and the strategies users develop anyway [106, 111]. Zong et al. then gave the area its clearest design vocabulary: structure (how a visualization’s entities are organized for traversal), navigation (the operations that move through that structure), and description (what is spoken at each position), showing through co-design that screen reader users want rich, structured exploration rather than a single summary [145]. Olli operationalized part of this insight as an open-source adapter that renders structured traversal for existing charting libraries [14]. A small number of production systems, notably Highcharts, SAS Graphics Accelerator, and Visa Chart Components, have shipped meaningful accessibility functionality [56, 104, 126]; they remain exceptions, and Chapter 4 examines that practitioner tooling landscape in detail.

Across all four modalities the same pattern recurs: each began with converting visual output into another channel, and each has progressively rediscovered that access is not a conversion problem but an interaction problem, turning toward structure, navigation, interactivity, and user control. The research itself has shifted accordingly: where early decades produced perception studies and one-off substitution systems, the last several years have produced co-designed interactive systems, design dimensions, and authoring environments. That convergence is part of why we argue, later in this chapter, that the remaining hard problems are not modality problems at all.

2.2.2 The Split Between Research and Practice

Before turning to the critical literature, it is worth naming that research and practice participate in accessibility at very different speeds, through very different methods, and with very different artifacts.

Accessibility *practice* organizes itself around standards. The Web Content Accessibility Guidelines carry legal and policy weight across much of the world [127], and a profession of accessibility specialists exists largely to translate such guidelines into concrete implementations. One example of this work would be ensuring that interactive elements carry functional semantics that assistive technologies can interpret, which is a foundational, simple, and necessary requirement for accessibility [128].

Standards are powerful precisely because they are consensus artifacts, or *politically mediated documents*, and for the same reason they are inherently reactive: developing, vetting, and formalizing them takes years, so they chronically trail the technologies they govern and are subject to intervention by individuals and organizations. Interactive data visualization sits at a painful point in that lag. Charts are semantically dense, interaction-heavy, and rendered through technologies the guidelines were not written for, so a practitioner who consults the standards in good faith finds little that speaks to their actual artifact. Robust evaluation in practice therefore leans on expert judgment and, in the best cases, direct involvement of people with disabilities, both of which are scarce resources [97].

Technical accessibility and assistive technology *research* meanwhile (outside of the previously mentioned work on modalities and our empirical record), moves quickly and publishes minimally-viable prototypes. These prototypes are an artifact practice largely cannot consume because it requires technical translation and methodological generalization from the small scope of an artifact into a system-wide implementation [31]. A practitioner cannot install a paper; a paper must be read and the artifact carefully dissected before being reconstructed or replicated. The result is a conversation in which researchers accumulate evidence and demonstrations on one side, practitioners wait on standards that lag by a decade on the other, and very little crosses between them. This split is not unique to visualization, but visualization makes it unusually visible, and closing it from the tooling side is the project of this thesis.

2.2.3 Empirical Depth, but a Lack of Applied Work in Production

What exists, first, is empirical breadth. Surveys and meta-analyses have mapped the research space and its biases: accessibility research broadly concentrates on blindness and on computer output [79], and visualization accessibility research mirrors that concentration, with vision-related work dominating while motor, cognitive, neurological, and vestibular access remain nearly unstudied [73, 86, 137]. Even within the well-studied territory the breadth has a shape: blind and low vision people are routinely researched as one population despite using different assistive technologies and different interaction practices [121], and the prototypes the field celebrates are overwhelmingly built for screen reader interaction in particular.

Second, the field has produced exemplary prototypes. The systems surveyed above, such as VoxLens, Chart Reader, Olli, and Umwelt, are rigorous, validated, and often co-designed with disabled collaborators.

What barely exists is the other half of the pipeline: practitioner tools in production environments, and practitioner action at scale. Studies of deployed visualizations and of the people who build them find that accessible outcomes remain rare and that practitioners struggle even when motivated, gathering fragmented guidance themselves with little way to judge its quality [67, 113]. Mainstream visualization tooling has invested its enormous momentum elsewhere, in expressiveness, scale, and interactive speed [15, 52, 105], with accessibility absent from core abstractions and left to individual developers, libraries, or projects. Research prototypes, meanwhile, rarely become infrastructure: toolkit research is evaluated primarily by demonstration [75], and most demonstrated systems are archived with their papers.

This asymmetry is the load-bearing observation of this chapter: **the field has a breadth of empirical work, yet it lacks practitioner tools in production environments and practitioner**

action. A practitioner today inherits strong evidence about what good access looks like but material change remains far behind. The natural next question is to investigate why this is the case, and whether and how tool-making can play a role in practitioner action.

2.3 Critical Voices and Work Along the Periphery

The second register of the conversation is critical, and although it is sometimes treated as peripheral to systems research, it sets the terms under which systems research should be read. We treat it here as load-bearing. This section will be brief, but essential.

The disability rights movement crystallized its central demand in the principle “nothing about us without us:” people with disabilities must hold power in the decisions and designs that concern them [20]. HCI and accessibility research have been working through what that demand means for our field’s methods, authorship, and self-conception [57, 116, 119, 120], including calls for a crip HCI and crip visualization that engages disability on disabled people’s own terms [60, 135] and concrete practices such as crediting disabled participant-contributors by name [144].

Several concepts from this literature directly shape this thesis. Mingus names *access intimacy*, the felt sense of someone genuinely getting your access needs, to insist that access is relational rather than merely technical and logistical [91]; Bennett et al. build on that disability justice lineage to propose *interdependence* as a frame for assistive technology research, countering the field’s fixation on independence [8]. Hamraie and Fritsch’s crip technoscience manifesto insists that disabled people are experts and designers of everyday life, and re-frames access not as a finished state but as friction, something contested and negotiated [49]. Hamraie’s history of universal design develops *access-knowledge*, showing that what counts as access has always been produced through political struggle rather than neutral technical progress [46, 47]. Bennett et al.’s biographical prototypes push against savior narratives in design by recognizing the prototyping disabled people already do in their own lives [9]. Shew names the broader cultural failure *technoableism*: the belief that technology will “fix” disabled people, held mostly by people who never asked disabled people what they want [114]. And within visualization specifically, Hsueh et al. bring crip technoscience to bear on data experiences, articulating how *access friction* arises when one design that includes some people excludes others [60].

We read this literature first as a set of constraints on what this thesis may claim. It is why we reject techno-solutionist framing outright: no tool in this thesis “solves” accessibility, because access is negotiated among people, institutions, and policy, and tools do not meaningfully redistribute power entirely on their own. It is also why access friction appears throughout this thesis as a design reality rather than an edge case, and why personalization is treated in Chapter 8 as an ethical question and not merely an intellectual problem to solve. And this body of work imposes a boundary we want to name early: a thesis about practitioner tooling mostly serves practitioners building *for* disabled users. The later chapters, Cross-perception and Softerware, begin to shift who holds the power to shape our interfaces. The critical literature is the measure of how far there is still to go.

We also read this literature as relatively divorced from accessibility work in practice. Critical scholarship has even been critical of “access” as a framing and its shortcomings [46, 47, 48, 97, 135]. Yet, practitioners are building systems all over the world that must be made accessible.

And rather than argue that people with disabilities should not be included in the design and development of new projects, instead we argue that, at least in some cases, the empirical record of what people with disabilities want and need is already enough imperative for builders and designers to begin working. For something new and unexplored? Yes, people with disabilities should be included. But what if we have a sufficient breadth and depth of guidance from people with disabilities already? Aren't we then ready to take action and begin practical application? *Chartability* in Chapter 4 takes this stance: people with disabilities have already said enough for us to build out comprehensive guidelines and an evaluation framework. Between standards, research, and community practice, we have an authoritative set of criteria we can test against. This should be our starting point; the bare minimum. And *Data Navigator* in Chapter 5 didn't seek to build a novel interaction but rather build a better system than what currently exists, following existing empirical work that laid the foundation for rich navigation as a necessity [145]. And in Chapter 5 we *do* demonstrate that our system enables novel interaction designs and how practitioners can approach co-design in a simple, effective, and thoughtful way with partners with disabilities. Chapter 6 sought to investigate the barriers that sighted practitioners (without disabilities) face, which is outside of the scope of critical disability literature, and Chapters 7 and 8 then centered on the experiences of blind and low vision people, and how they might have interactive and innovative power over their experiences.

2.4 Why Navigation, Interaction, and Personalization

If the gap were uniform, any contribution would do. It is not uniform. Looking across the empirical record, the prototypes, and the practitioner studies, the places where practitioners most consistently lack the means to act cluster in three domains, and each domain fails for a different kind of reason.

It is worth being explicit about why these three and not others, because the field's other major needs are of a different kind. Description is the best-served domain: practitioners have models, guidance, and concrete heuristics for what to write [68, 77], and what remains is largely a matter of doing it. Evaluation is hard, but it is hard as a knowledge problem, a matter of gathering, judging, and applying scattered guidance, and a well-built resource can carry a practitioner a long way; Chapter 4 contributes exactly such a resource and uses it to map the territory. *Chartability*, from Chapter 4, ultimately ends up framing the rest of this thesis. As we did work alongside our co-designers, clients, and industry partners over the years, applying *Chartability* to their work, it became clear that three major barriers emerged as the most pernicious (or least-resourced) for practitioners to address.

Navigation, interaction, and personalization are different in kind: in each of them, a practitioner who knows precisely what good access requires still cannot build it, because the means do not exist in their tools. Guidance cannot substitute for means. These areas then serve as opportunities to innovate, to build better, more general, or more specific tooling.

We unpack these three tooling gaps more below.

2.4.1 Navigation

The empirical case for structured navigation is now strong. Zong et al. showed through co-design that screen reader users want to traverse a visualization’s structure, not only hear a summary: structure, navigation operations, and description together determine the quality of the experience [145]. Systems work has begun to deliver this for specific stacks, with Chart Reader designing navigation experiences with screen reader users [124] and Olli adapting existing charting libraries into traversable structure [14]. What practitioners still lack is twofold. They lack low-level building blocks: a way to give *any* data interface, whatever its renderer or framework, a navigable structure with arbitrary shape, rather than the fixed hierarchies particular systems assume. And they lack any visible, manipulable representation of navigation structure in their workflows: a sighted practitioner can see a color palette and act on it with existing tooling but cannot see a navigation graph and act on it, which makes navigation design incredibly burdensome to iterate on and wholly downstream from other visual work. Chapter 5 (*Data Navigator*) contributes the structural substrate, and Chapter 6 (*Skeleton*) contributes the authoring representation; their chapter-level related work sections cover visualization toolkits and authoring systems more in depth.

2.4.2 Interaction

History shows that input design, not just output conversion, changes what is possible. Kane et al.’s *Slide Rule* contributed multi-touch techniques that let blind users operate touchscreens, outperforming a button-based baseline and prefiguring techniques that later appeared in commercial screen readers [69]. That simple lesson has not yet reached data analysis. Accessible visualization interaction today is overwhelmingly what we call *access-oriented*: it exposes information that is already there, through navigation, sonification, or description. These methods are *interactive*, as existing work as argued [145], but we argue that the interaction is focused on perception of the representation and not enabling analytical interaction with and manipulation of the information that lies beneath the representation. Analytical work of the kind sighted analysts perform with cross-filtered, coordinated views depends on rapid, continuous loops of input and perception [52, 76], and a screen reader’s serial, query-at-a-time interaction may be structurally unable to sustain such loops. Fast, complex manipulation of a data space is presently made inaccessible by the primary assistive tool used by blind analysts. Rethinking input for blind analytical interaction, beyond an *access-oriented* framing, is nearly unexplored territory. Chapter 7 (*Cross-perception*) takes this on, formalizing a design framework and contributing a hardware prototype; that chapter’s related work covers tactile and haptic input devices in more detail.

2.4.3 Personalization

Accessibility research has long argued that systems, not users, should do the adapting. Put simply, this means that systems must be capable of adaptation. Arguably, this is the “soft” part of *software*. Yet, this softness can create conflict in system design, especially when an entire field of work must refactor their dominant tooling in order to enable system adaptation by end users. Visualization faces this exact challenge.

Existing work has demonstrated that interfaces adjusted to fit a person’s abilities and preferences measurably improved speed, accuracy, and satisfaction for users with motor impairments compared to default interfaces [34], and ability-based design generalized the stance into principles [139]. Good systems are made to fit a user’s abilities and remove barriers that they face. It could be argued, in broad strokes, that the field of HCI centers on the goal of *fit* between a user and a computational system.

Yet, visualization toolkits are largely not made to be personalized by end users. They are rigid and *hard*; lacking that softness required for end-user manipulation. Visualizations are typically built with a *universal design* [47] approach, when being made accessible (if being made accessible at all): one design is intended to fit as many people as possible. In visualization, recent work shows screen reader users benefit from customizing even small things, like the textual tokens spoken during navigation [66]. Yet a significant gap exists that enables visualization systems to adapt to personalization at scale and for many different people. How should these systems be built and what would they require? Which dimensions and features and functionality of data visualizations *should* be manipulable and customizable to fit end users? How should systems be constructed in order to make personalization effective and useful? And ultimately: how do we make visualization toolkits *softer*?

Some existing work on customizable accessibility settings identifies what drives people to adopt or abandon personalization features [141], but not all of this work neatly translates into visualization system design and development. Additionally, and fundamentally, we simply lack an empirical record of what exactly different people with disabilities would even want or need to manipulate. Presently, no mainstream visualization library or system offers practitioners infrastructure for end-user preferences over data representations: no shared option vocabulary, no persistence, no way to reconcile preferences with authorial defaults. Most visualizations become completely static in their design. Those with manipulable elements are only manipulable through hacking or manual overrides, placing a significant burden of access on end users, exactly where ability-based design says it should not [139]. Chapter 8 (*Towards Softerware*) engages this domain in collaboration with a production charting library, considered by the empirical record to be the most-accessible toolkit according to present evaluation methods by a wide margin [74]; Chapter 8’s related work section covers adaptive systems and personalization research in more depth.

2.5 What to Take Away

Five things from this chapter frame everything that follows. First, tools direct the work and the workers who use them, which makes tool-making a rich potential site for accessibility intervention and research into better understanding practitioner barriers when making and using tools. Second, the conversation between accessibility and data visualization is old, rich, and increasingly self-critical. Third, the field’s output is lopsided: empirical breadth, exemplary prototypes, and guidance exist in abundance, while production tooling and practitioner action barely exist. The broader field’s critical voices impose real constraints: tools do not solve access, and this thesis will not claim they do. Instead, critical work implores us to bridge the gap between research and practice. People with disabilities have been saying enough, it’s time to get to work

and try to apply what we know at scale. Fourth, the gap is concentrated where practitioners' means run out in three specific ways: navigation lacks a structural substrate, interaction lacks input techniques for analysis, and personalization lacks infrastructure. Fifth, the first step in our thesis work should be to collect the empirical record and synthesize it for applied accessibility work in the context of interactive data visualization. This is Chapter 4's *Chartability* project. Given where the field stands, *navigation*, *input*, and *personalization* are the categories of work that follow Chapter 4; the chapters ahead carry this out as the body of this thesis, and our concluding chapters return to what tool-making accomplished and what it cannot accomplish, on the other side.

Chapter 3

Overview of Contributions

In this dissertation, we contribute practical advancements in tool-making as an intervention on the accessibility of interactive data experiences. The thesis of this dissertation is as follows:

This dissertation argues that the tools practitioners use to build interactive data experiences are themselves sites where accessibility barriers are produced, prevented, or alleviated for both end users and authors. This work contributes five tools, Chartability, Data Navigator, Skeleton, Cross-perception, and Softerware, that collectively center accessibility work on the empowerment of disabled and non-disabled practitioners across the full arc of evaluation, data navigation, analytical interaction, and personalization.

Rather than framing accessibility research solely around ideal experiences for end users with disabilities, this thesis investigates why accessibility work is so difficult for the practitioners who build interactive data experiences and what tool-making can reveal about those difficulties. We organize this investigation around four domains where practitioners face the most persistent challenges: evaluation, navigation, interaction, and personalization. *Chartability*, a heuristic framework, contributed first and mapped the full landscape of accessibility barriers. *Chartability* helped us to identify and prioritize our three latter domains (navigation, interaction, and personalization) as the areas where the most severe gaps remain in practitioner work. *Data Navigator* and *Skeleton* then investigate navigation, finding that practitioners struggle because navigation structure has no visible, manipulable representation in their workflows. *Cross-perception* engages interaction, demonstrating that blind data analysis has been constrained by existing tools and that a new interaction design framework can reshape what analytical work is possible. *Softerware* addresses personalization, revealing that access needs genuinely conflict across users and that meaningful personalization requires system-level infrastructure that does not yet exist in practitioner tooling ecosystems.

We engage each domain below with the questions: “Why does this work matter?”, “Why is it hard?”, and “What has tool-making within this domain showed us?”

The four domains of work that we engage in this thesis start first with **evaluation**. In work contexts where someone is designing and developing interactive data experiences, the practitioner must have the knowledge, tools, and resources available to systematically identify how their interfaces produce barriers for people with disabilities. A significant portion of professional accessibility work (arguably most, if not all) is founded on auditing and evaluating barriers to access. This work is pre-dominantly done through a standards-based approach [127], although in more robust evaluation work, people with disabilities are actively involved in the process [97].

Evaluation is difficult work because much of it is contextually defined by the author themselves, and most tasks at this intersection require careful, non-automated processes and meth-

ods [25, 97]. To make matters more difficult, no comprehensive guidelines, tests, and tools exist in any singular location. Practitioners often must gather these resources themselves, which tend to be situated towards accessibility in general or are high-level and provide minimal usefulness in practice. Additionally, practitioners themselves often have little knowledge about the veracity or quality of any given bit of information they gather [67, 113], and often do the work themselves to synthesize this disparate space of information into a usable format they can apply to their own evaluation work.

To engage this, our first main chapter focuses on *Chartability*, a heuristic framework that enables designers, developers, and auditors to systematically evaluate data visualizations and interfaces for a wide range of accessibility barriers, considering people with visual, motor, vestibular, neurological, and cognitive disabilities. In this project, we did the hard work for other practitioners and contributed our collection of synthesized accessibility resources in a single workbook. We had practitioners try out our resource in real environments, in-situ, in order to learn more about the challenges and barriers they themselves faced in evaluation work. With *Chartability*, practitioners, especially those with limited accessibility expertise, gained more confidence and clarity in assessing and improving their work. Additionally, *Chartability* has since become widely applied as a framework that isn't just used for evaluation but also as design guidance in many contexts, internationally, including policy organizations, governmental groups, and more than 100 companies and businesses.

Chartability then opened up a significant landscape of new projects and research directions. From a combination of my existing expertise as a visualization designer and engineer, in addition to our continued application of *Chartability* in the wild, we began to identify the trickiest and most-difficult domains of work for practitioners. *Chartability* has 50 total heuristics, or tests, each organized under one of seven principles. These three domains did not simply “emerge” on their own; they were evidenced by our applied work. Alongside the research activities that comprise this thesis, in a freelancing role we applied *Chartability* by partnering with industry, government, and non-profit organizations, conducting over 150,000 tests (both manual and automated) and working closely with dozens of designers and engineers. From that body of work, three larger domains stood out as the areas where the most severe and dramatic accessibility barriers remained unaddressed: data *navigation*, analytical *interaction*, and interface *personalization*.

So the next section of this thesis engages the first of these three: **navigation**. Navigation is a fundamental type of interaction that is leveraged by modern software-based assistive technologies. Screen readers, the primary tool used by people who are blind to interact with computers, navigate content. Additionally, many other assistive technologies, such as a sip and puff device (like the “POSSUM” from as far back as '63 [82]) also navigate. Navigational technologies are leveraged by people with a significant array of disabilities, yet tend to be entirely ignored by existing data visualization tools, which are pre-dominantly built to support direct input (using a computer mouse).

Empirical work has already demonstrated that structural navigation is actually good [145], even when regular alternative text (image descriptions) exist. This is both because people who are blind can gain both a high level understanding (from the description) as well as lower-level sense of the data's structure and arrangement, in addition to the fact that discrete, structural navigation exposes interactivity that may exist on any visualization elements (such as they can be hovered or clicked with a mouse in order to perform some action). So, if good empirical work exists: *why*

haven't practitioners put this research into action? What makes this work hard to do?

We first built *Data Navigator* to provide the building blocks we needed in order to address the technical and conceptual gaps that were required to make any visualization or visualization tool provide a navigable, interactive structure. *Data Navigator* is a low-level toolkit which can be used to construct accessible navigation structures such as lists, trees, and diagrams from an underlying graph structure. We leveraged graph theory for an applied HCI problem: nodes and edges represent any relationships within the data as a structure, which then supports rich expressiveness of data navigation experiences. Users can navigate discrete marks in a visualization, clusters, groupings, and more. In addition to its structural scaffolding, *Data Navigator* also supports a wide array of input modalities leveraged by people with disabilities (screen readers, keyboards, speech, and gestures).

Data Navigator provided a substrate, but this contribution alone wasn't enough to engage the question *why is navigation so hard, in practice?*. So the chapter following *Data Navigator* introduces *Skeleton*, a data navigation authoring tool built on top of *Data Navigator*. Our novel approach in *Skeleton* involves visualizing and making manipulable the nodes, edges, and textual data that comprise non-visual end user experiences. *Skeleton* visualizes the building blocks that comprise *Data Navigator*. Additionally, *Skeleton* provides expressive, rapid scaffolding capabilities that leverage data visualization rendering engines. This scaffolding engine helps practitioners quickly create common configurations for non-visual data navigation structures that retain visual congruence to the underlying structure.

But most importantly, *Skeleton* serves as a framework that shapes designerly consideration. Our conjecture was that because sighted practitioners cannot *see* navigation building blocks, they will not treat those elements as iterable design materials. We conjectured: Navigation is hard in practice because sighted designers face barriers to iteration and understanding. We conducted an empirical study with sighted practitioners and found that making non-visual elements visual helped practitioners shift from treating accessibility as a compliance task to treating it as a design problem, re-iterating on the visual aspects of their design, and engaging in the complex and nuanced components that comprise data navigation experiences.

Now, **interaction** becomes the next area we want to engage. Existing accessible data interaction for people who are blind, including our previous work on navigation (which is a form of interaction), predominantly seeks to expose information. This is what we call *access-oriented interaction*. In terms of low-level analytical tasks, most are then made feasible through navigation, sonification, or summarization-based and question-answering approaches: retrieving values, filtering, computing derived values, sorting, determining ranges, clustering, and finding outliers. What remains are analytical tasks that, despite being “low level” (understood as *unable to be reduced into more fundamental tasks*), are cognitively highly complex: finding correlations and characterizing distributions [3]. These tasks require complex hypothesization and exploration, rather than a system that simply encourages surfacing what is known or what is already present in the data: it requires combining, remixing, restructuring, and dividing data.

But blind *analytical interaction* isn't just important to engage because it is understudied, it is important to engage because many of interactive information visualization's most impactful tools for data science enable it [32, 52, 94, 130]. In visualization, *cross-filtering* is one example of an interaction that enables a user to filter one visual space while seeing a coordinated change in another visual space simultaneously and near-instantaneously. The speed of input interaction

and perception of output also matters: even a small bit of latency changes the quality of a user’s data exploration activities [76]. We conjectured that a screen reader, the most-used tool leveraged by blind people when interacting with computers, may be insufficient for engaging this task.

To engage this, this thesis introduces *Cross-perception*, an approach for building analytical interactions that support perception in one space of input interaction with simultaneous, non-competing perception of output in another space of data representation. We first formalized a design framework for producing *cross-perception* experiences and then built a novel prototype device, the *cross-feelter*, that enables blind *cross-perception* of a cross-filtering data exploration interface. In an empirical study with blind users (with and without existing data expertise), we found *cross-perception* speeds up analytical exploration by 90% and helps blind users consider vastly more questions of their dataset (+188% computational queries, +54% spoken aloud) compared to a screen reader-driven interaction.

Beyond performance, we found that our input modality itself shaped the character of analytical engagement: participants didn’t just work faster, they considered more dimensions of their data and asked qualitatively different questions. The *cross-feelter* also reduced anxiety and substantially increased enjoyment, particularly for participants without prior data expertise, suggesting that the barriers blind practitioners face in data work are not only functional but affective. Additionally, we had our blind practitioners imagine new interaction possibilities that *cross-perception* could enable including and beyond our *cross-feelter* device.

Our final domain of work is the most difficult for visualization practitioners to engage: **personalization**. While *navigation* demands better software tooling and visual support and *interaction* requires new hardware, *personalization* completely re-orientes how software authoring takes place. Personalization matters because of *access friction*, which is a design challenge where one design or interface configuration that might be accessible for one person or group of people turns out to create barriers for someone else [46, 60]. In existing work on personalization and accessibility, studies have demonstrated that end-user control is great to have and can alleviate friction [66, 141], but little work has been done to explore what personalization looks like for an existing data visualization library and how practitioners should build and maintain their existing systems to support it.

Our final chapter introduces *Software* to address the tension between standardized accessible design and the diverse needs of real users with disabilities. In the wild, *access friction* exists in every design that reaches a public audience; it is inevitable. This tension ultimately means that some users have a worse experience, and may even face exclusion, with any particular design configuration. So for this work, we conducted our research in-situ with visualization software engineers and designers and worked to build a scalable, flexible software system dubbed “softerware” that enables end users to manipulate the appearance and functionality of the charts and graphs they encounter according to their own preferences. We conducted empirical research to inform our collaborators, as well as other visualization system authors, with guidelines and considerations for building *softerware* systems. In our study, no two participants chose the same preference configuration and participants with the same diagnosed condition sometimes needed opposite design treatments. Practitioners, meanwhile, immediately raised ethical concerns about whether personalization would let designers off the hook for poor defaults. These findings revealed that the real barriers to personalization are not at the level of any individual chart but at the level of system infrastructure: without persistence, cross-system interoperability, and shared

standards, the effort required of end users exceeds the value they receive.

Combined, these contributions provide empirical insights and practical advancements in the state of the art for tooling that bridges gaps in current accessibility practices in visualization and data science. Our work ultimately enables people with and without disabilities to better evaluate barriers in, analyze with, design for, develop, and personalize interactive data experiences. We demonstrate that tool-making is a productive intervention that both engages accessibility barriers and elucidates why those gaps exist in practitioner work.

Part VI

Conclusion

Chapter 9

Findings

The five projects in this thesis were presented as five chapters, each with its own results and its own conclusion. Read separately, they contribute a heuristic resource for evaluation, a structural toolkit for navigation, an authoring environment for non-visual design, an interaction technique and device for blind analysis, and a system and study of end-user personalization. This chapter states what they show *together*. Chapter 2 closed on a lopsided field: a breadth of empirical work and exemplary prototypes on one side, and a near absence of practitioner tools in production environments and practitioner action on the other, with the hardest remaining challenges concentrated in navigation, interaction, and personalization. The findings below are what this thesis learned by working inside that gap.

Each finding is a synthesis across chapters, and each is traceable to results reported in the body of this thesis. We have written them for two audiences at once: practitioners who build, evaluate, and maintain data experiences, and researchers who study how that work happens. Where a promising idea outran the evidence, it is not here; it is in the next chapter, with the rest of the discussion and future work.

9.1 Access Barriers Are Produced for Practitioners, Both With and Without Disabilities

The social model holds that disability is not located in a body but produced in the interaction between a person and a human-made system [85]. This thesis began by applying that lens to end users: visualizations, like curbs, produce barriers. The cross-cutting finding of this thesis is that the same lens applies, with the same force, to the practitioners doing accessibility work, and that the materials of that work are among the systems producing the barriers.

Every project surfaced a version of this. In *Chartability*, the guideline landscape itself was a barrier: standards and research were scattered, often high-level, and difficult to apply to complex interactive data interfaces, and practitioners had little way to judge the quality of what they gathered [67, 113]. Guidelines written to remove barriers for end users limited what practitioners could reasonably understand and act on, and an expert visualization practitioner in our study admitted he did not know where to start. In *Skeleton*'s co-design foundations, *Data Navigator* made better end-user experiences possible, but because its structures were entirely non-visual and code-bound, it produced severe iteration barriers for sighted practitioners: they could not perceive what they were building, and prototypes were misinterpreted or abandoned. And in *Cross-perception*, the screen reader, the most important enabling technology for blind computer use, also limited what blind practitioners could reasonably do analytically: its serial, discrete traversal kept participants in an access-oriented mode that structurally separated interaction from information, with measurable costs in time, completion, and the number of questions they could ask of their data [111].

The operationalized version of this finding: when accessibility work stalls, look for barriers

in the practitioner’s tools and materials before attributing the failure to motivation or knowledge. None of the practitioners in these studies lacked care. What they lacked, in each case, was a working relationship with materials that had been shaped without them in mind, and this is true whether the practitioner is sighted, blind, disabled, or not. The same observational stance the field has learned to take toward end users belongs in the study of the workers, too.

9.2 Tools Can Reduce the Work Required to Make Things Accessible

Chapter 2 argued that guidance cannot substitute for means. The constructive half of that argument is that supplying means measurably changes what practitioners can do, and three chapters provide the evidence.

Chartability reduced the perceived difficulty of evaluation and increased the confidence of practitioners new to accessibility, whose audit results were reasonably comparable to the authors’ own; beyond the study, it has been applied by policy and governmental organizations and more than 100 companies, across toolchains from Tableau and Excel to D3 and Figma. *Data Navigator* gave previously inaccessible rendering formats a navigable structure: a static raster image gained full multi-input navigation with no automatically detectable conformance failures, and a deterministic ingestion pipeline recreated, and then improved on, an existing toolkit’s navigation scheme for rendering modes that previously had none. *Skeleton*’s scaffolding collapsed the spatial authoring of a navigation structure to one or two minutes of refinement, and within single sessions every participant iterated substantively on designs that, in a code-only workflow, our co-design collaborators had struggled to iterate on at all.

None of this is evidence that tools solve access; the boundaries of that claim get their own finding below. It is evidence of something narrower and more useful: the labor of access work is sensitive to tooling, and reductions in that labor are achievable, measurable, and repeatable across very different kinds of tools, from a workbook of heuristics to a rendering-independent toolkit to an authoring interface.

9.6 The Hardest Barriers Required Intervention at the Substrate

Chapter 2 characterized navigation, interaction, and personalization as a missing structure, a missing technique, and a missing infrastructure. Having now worked in all three domains, we can state what they share: in each, the barrier sat one level below where practitioners are ordinarily able to act, in the substrate their work is built on, and in each, outcomes changed when the intervention went to that level rather than adding features above it. This finding is held heavily in tension with existing research that tends towards higher levels of software and interaction abstraction, not lower. A massive wave of new research has emerged that essentially asks, “what new layers or wrappers can we create around existing interfaces and user experience problems, using large language models and agents?” While our thesis doesn’t engage the effectiveness or

intellectual value of those projects, instead we hold this thesis in fundamental contrast to their approach: that some of our most difficult gaps are best engaged below our present levels of software abstraction, not above.

The evidence is consistent across the three parts of this thesis. No amount of feature work on top of canvas-rendered pixels can recover semantics that the rendering substrate never carried, and forcing complex, non-hierarchical data relationships into HTML’s hierarchical substrate limited what was possible to build. This is precisely what a large and passionate resistance to “overlay” technologies engages within both the practical and academic accessibility communities [28, 42, 50, 81]. There may simply be a limit to how much good data, if at all, can be generated from bad or missing data [13, 96].

Data Navigator changed outcomes by supplying a structural substrate of its own, a graph independent of any renderer. This requires underlying data that is supplied with sufficient quality, what Doug Schepers calls an “underlay” rather than an overlay [107].

No software feature on existing consumer hardware approximates persistent physical state and simultaneous perception of input and output, the properties analytical interaction depends on [76]; the *cross-feelter* changed the hardware substrate, and the character of blind analysis changed with it, from accessing data toward interrogating it, at nearly three times the query rate.

And per-chart personalization features fail the effort-to-outcome test that end users themselves applied; what every participant and practitioner in the *Software* study named instead were system-class needs [31]: persistence, profile sharing, interoperability, and standards beneath any individual chart.

This is why the substrate matters as a space of intervention: in all three domains, the ceiling on what a motivated, well-informed practitioner could make accessible was set by material beneath their reach, so guidance, effort, and even exemplary prototypes hit that ceiling without moving it. The work in this thesis moved outcomes exactly where it changed the material itself, and the same evidence shows the stages of this work leaning on one another: structure without authoring support went unused, authoring raised evaluative questions practitioners could not answer alone, and personalization without accessible defaults and persistent infrastructure shifts burden onto the people it is meant to serve. Researchers and toolmakers deciding where to spend effort in these three domains have, in this thesis, a consistent answer about where the leverage was found.

Chapter 10

Discussion & Future Work

10.2 “Applied” accessibility work and *low research*

I came into an academic research environment already as an engineer. I had questions to big problems, especially about how tools might shape human behavior and how we can use that towards ethically good outcomes. But *Chartability* was a project I made first for myself, to engage the problems I, personally, was facing.

And *Data Navigator* was motivated by my existing experience making visualizations navigable. I knew we had to build better tools, because continually trying to make a hierarchical substrate like HTML work for complex, non-hierarchical relationships simply limited what was possible and easy to do.

These two projects then led to many opportunities to test them in applied contexts. Our CZI grant was motivated towards the application of both, Adobe was interested in *Data Navigator*, and the University of Wisconsin’s GIS folks wanted to figure out how to make a complex map navigable, *Data Navigator* or not. Yet, as a researcher I wasn’t incentivized to pursue these projects. In fact, they all proved to have no immediately apparent novelty. It was a risk to spend my time in these spaces.

Applied work is structurally and socially devalued in academic research environments, seen often as a lower form of research and design compared to more “pure” forms of knowledge production. (We conjecture that in part this is due to the scope of knowledge that practitioners produce: they work to take broad knowledge and apply it to specific, small problem spaces. Foundational research tends to assume that broad knowledge does not then bubble up from this work, that it is a one-way flow.)

In my first year as a PhD student I was given the advice by a senior researcher not to hold my experience as a practitioner too highly because, “practitioners often don’t know what they are doing.” But I did, of course, to some degree. Yet, this attitude is pervasive in some spaces and is a problem because it limits what kinds of knowledge we care to attend to.

Initially, I felt that with *Chartability* I had lucked out. But after four more collaborations which resulted in funding and two research publications, we can now argue that this method has been reliable. In *Softerware* and *Skeleton*, pursuing practical problems in applied environments *did* serve as a fruitful source of generalizable knowledge and advancements in the state of the art. But these projects, in addition to *Chartability*, were all motivated by immediate, non-novel problems: people knew their charts were inaccessible and wanted some design guidance or engineering solutions.

So perhaps “low research” works.

Yet, some research communities, especially *ASSETS*, have yet to reckon with “applied” work like this. If you trace the lineage of *action research* from Hayes [51] (which is how we frame our applied work in *Skeleton*), it leads into fantastic areas of participatory and community-based work, often with or by marginalized people and people with disabilities [8, 60, 78, 83, 116, 119, 120, 125, 136]. We would even argue that this lineage of work, which prioritizes and centers the

lived experience of a person, is also why we have seen so many successful auto-biographical and auto-innovation styles of research as well recently [12, 35, 36, 57, 109, 134, 135].

And yet, action research largely does not touch on how practitioners (particularly those without disabilities) do applied work in accessibility [67, 113]. We believe that significant gaps exist in the space between traditional “high” knowledge production and applied, “lowly” spaces. Researchers tend to pursue novelty and the lifecycle of research production is highly risk averse. And yet, relatively little work has been done within the accessibility research community to intellectually address the actual social and cultural risks involved in producing decades of useless, lost, and forgotten prototypes and design guidelines [61, 63]. We would conjecture that research-producing academics are overly concerned with academic readership relative to the myriad of ways that their knowledge production actually is used by the rest of the world [92]. We would challenge *ASSETS* and the AT/accessibility communities of *CHI* in particular to take seriously the lack of applied (industry, government, and non-academic) work it produces as a community.

10.3 Who is responsible for repair?

Lastly, we want to revisit one of this thesis’ opening points, where we argue that the *tool-makers* are first responsible for repair. This is true. However, the most pressing issue we have faced in recent years is mostly unmentioned across these research projects: tool-makers might be responsible, but this is because they are the only ones who have the *power* to make things accessible. Does this always need to be the case? Can we imagine an artifact’s authority over the user’s ability to access being designed towards self-subversion [39] or de-centralized agency [19, 71, 93], instead? What might that look like, concretely?

In *Software*, we begin to engage this larger problem in terms of an idealized state where a user can repair or re-design their own experiences. But to me, this self-repair is like laying down train tracks for yourself as you move a locomotive, but then lifting up your own tracks behind you as you go. You’re the only one helping yourself. This is not ideal, for you or others.

What we need are broad, lasting, infrastructural changes. On the web, this problem becomes quite difficult to solve. A personal computer or device? Again, someone can auto-design their interfaces into a better state. But back when we started *Chartability*, the WebAIM Million’s report showed more than 95% of the top one million website home pages contain at least one critical accessibility error. And now, more than six years later, that proportion is unchanged [131].

Some had imagined that generative AI would solve the massive infrastructural repair problems we face. But unfortunately, the latest WebAIM Million report shows that since 2020, ARIA usage has increased and correlates to more errors, while use of tabindex on a page has increased nearly 300% and also correlates to more errors on a page. If anything, during the age of generative AI, we have seen existing bad patterns worsen in prevalence and complexity.

We firmly believe that a tools-based approach is not enough on its own. Tool-making cannot be the *only* intervention on inaccessibility. Tools and tool-making, as our thesis argues, have a powerful role to play. But we simply can’t tool our way out of failed infrastructure and inadequate policy when someone else *owns* the tools and tool-making. Visiting a website is like going into someone else’s home: arranged according to their effort, tastes, and so on. If you can’t access their home, you essentially need to request that they let you in personally. Website repair always

falls to the owner and maintainer of a website, and they largely don't take any meaningful action.

Sidewalks outside of homes are a good parallel to this problem. Sidewalk accessibility is a massive infrastructural problem [101], and yet localities treat sidewalk maintenance in different ways: some, like where I, Frank, presently live in the south hills of Pittsburgh, put the onus on the homeowner whose house and property the sidewalk touches. In other places, sidewalks are considered a public path, similar to a roadway, and are maintained through public tax and resource management. To no surprise, privately-maintained, public-access sidewalks are worse for people in pretty much every way than publicly-maintained ones [140]. This is because private homeowners don't care about sidewalk maintenance unless the city manages to fine them or they get sued.

And the web is a collection of private spaces that you visit privately. There is no truly shared, universally democratic, public space on the web. Centralization is partly to blame: sharing space while scaling leads to consolidation.

So our future work will continue to wrestle with the same tensions of scale, repair, and anti-consolidation of power, motivated by the same WebAIM Million report. But now we look to questions of *democratic* and *radical* access to accessibility repair. The barriers we hope to tackle in the future are political and infrastructural. Perhaps tool-making will participate in this work, but it seems clear now from our work that the upstream technical problems and socio-political conditions that tools inherit, will likely not be addressed by tools alone.

What does “democratic” and “radical” infrastructure work look like? It will probably be an extension of *Softerware*, to some degree. We imagine future research that explores public-first spaces, ones where access is socially negotiated and repair belongs to all of us. Is this an autonomous space, like an autonomous zone [11] separate from the web? Above it, looking down into it, like shared annotation tools but capable of sharing the manipulation of websites [100]? A space with ambient co-repair, modeled after projects that bring people together [110] or that allow community “fixing” of misinformation [64]? Perhaps feminist thought on the ethics of care can help us [53, 80]? Or maybe it will be something else entirely; I'm not yet sure. But what made the web fantastic years ago is long gone; most of it has been hedged into corporate spaces that are controlled, maintained, and repaired by corporate power. And these entities are notoriously bad at repair. What we imagine in the future involves reclaiming a sense that the web is *ours*, belongs to *us*, and that ultimately *we* are responsible for making it accessible.

References

- [1] Sara Ahmed. *Queer Phenomenology: Orientations, Objects, Others*. Duke University Press, Durham, NC, 2006. ISBN 9780822339144. ISBN: 978-0-8223-3914-4. [2.1.2](#)
- [2] Sara Ahmed. *What's the Use? On the Uses of Use*. Duke University Press, Durham, NC, 2019. ISBN 9781478006503. ISBN: 978-1478006503. [2.1.2](#)
- [3] R. Amar, J. Eagan, and J. Stasko. Low-level components of analytic activity in information visualization. In *IEEE Symposium on Information Visualization, 2005. INFOVIS 2005.*, page 111–117. IEEE, November 2005. doi: 10.1109/infvis.2005.1532136. URL <https://doi.org/10.1109/infvis.2005.1532136>. [3](#)
- [4] Apple Inc. Audio graphs. <https://developer.apple.com/documentation/accessibility/audio-graphs>, 2021. Apple Developer Documentation. Feature introduced at WWDC 2021. Accessed: 2026-06-12. [2.2.1](#)
- [5] Catherine M. Baker, Lauren R. Milne, Ryan Drapeau, Jeffrey Scofield, Cynthia L. Bennett, and Richard E. Ladner. Tactile graphics with a voice. *ACM Transactions on Accessible Computing*, 8(1):1–22, January 2016. ISSN 1936-7236. doi: 10.1145/2854005. URL <https://doi.org/10.1145/2854005>. [2.2.1](#)
- [6] BANA. Guidelines and Standards for Tactile Graphics. Braille standard, Braille Authority of North America, 2010. Accessed: 2021-09-03. [2.2.1](#)
- [7] Michel Beaudouin-Lafon, Susanne Bødker, and Wendy E. Mackay. Generative theories of interaction. *ACM Transactions on Computer-Human Interaction*, 28(6):1–54, November 2021. ISSN 1557-7325. doi: 10.1145/3468505. URL <https://doi.org/10.1145/3468505>. [2.1.2](#)
- [8] Cynthia L. Bennett, Erin Brady, and Stacy M. Branham. Interdependence as a Frame for Assistive Technology Research and Design. In *Proceedings of the 20th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '18, page 161–173, New York, NY, USA, October 2018. Association for Computing Machinery, ACM. doi: 10.1145/3234695.3236348. URL <https://doi.org/10.1145/3234695.3236348>. Accessed: 2021-09-03. [1.1.1](#), [2.3](#), [10.2](#)
- [9] Cynthia L. Bennett, Burren Peil, and Daniela K. Rosner. Biographical prototypes. In *Proceedings of the 2019 on Designing Interactive Systems Conference*, pages 35–47. ACM, 2019. doi: 10.1145/3322276.3322376. [2.3](#)
- [10] Dan Bennett, Alan Dix, Parisa Eslambolchilar, Feng Feng, Tom Froese, Vassilis Kostakos, Sebastien Leriche, and Niels van Berkel. Emergent interaction: Complexity, dynamics,

- and enaction in hci. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*, CHI '21, page 1–7. ACM, May 2021. doi: 10.1145/3411763.3441321. URL <https://doi.org/10.1145/3411763.3441321>. 2.1.2
- [11] Hakim Bey. *TAZ: The temporary autonomous zone, ontological anarchy, poetic terrorism*. Autonomedia, 1991. 10.3
- [12] Brandon Biggs, Christopher Toth, Tony Stockman, James M. Coughlan, and Bruce N. Walker. Evaluation of a non-visual auditory choropleth and travel map viewer. In *Proceedings of the 27th International Conference on Auditory Display (ICAD 2022)*, pages 82–90. International Community for Auditory Display, June 2022. doi: 10.21785/icad2022.027. URL <https://doi.org/10.21785/icad2022.027>. PMCID: PMC10010675. 2.2.1, 10.2
- [13] Abeba Birhane, Vishnu Boddeti, Sanghyun Han, Sasha Luccioni, and Vinay Prabhu. Into the laion’s den: Investigating hate in multimodal datasets. In *Advances in Neural Information Processing Systems 36*, NeurIPS 2023, page 21268–21284. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2023. doi: 10.52202/075280-0930. URL <http://dx.doi.org/10.52202/075280-0930>. 9.6
- [14] Matt Blanco, Jonathan Zong, and Arvind Satyanarayan. Olli: An Extensible Visualization Library for Screen Reader Accessibility. In *IEEE VIS Posters*, 2022. URL <https://vis.csail.mit.edu/pubs/olli>. Accessed: 2026-03-30, <https://vis.csail.mit.edu/pubs/olli>. 2.2.1, 2.4.1
- [15] M. Bostock, V. Ogievetsky, and J. Heer. D³ data-driven documents. *IEEE Transactions on Visualization and Computer Graphics*, 17(12):2301–2309, December 2011. ISSN 1077-2626. doi: 10.1109/tvcg.2011.185. URL <https://doi.org/10.1109/tvcg.2011.185>. 1.1.2, 2.2.3
- [16] Geoffrey C. Bowker and Susan Leigh Star. *Sorting Things Out: Classification and Its Consequences*. MIT Press, Cambridge, MA, 1999. ISBN 9780262522953. ISBN: 978-0-262-52295-3. 2.1.2
- [17] S. Brewster. Visualization tools for blind people using multiple modalities. *Disability and Rehabilitation*, 24(11-12):613–621, January 2002. ISSN 1464-5165. doi: 10.1080/09638280110111388. URL <https://doi.org/10.1080/09638280110111388>. 2.2.1
- [18] Jennifer L. Burke, Matthew S. Prewett, Ashley A. Gray, Liuquin Yang, Frederick R. B. Stilson, Michael D. Covert, Linda R. Elliot, and Elizabeth Redden. Comparing the effects of visual-auditory and visual-tactile feedback on user performance: a meta-analysis. In *Proceedings of the 8th international conference on Multimodal interfaces*, ICMI06, page 108–117. ACM, November 2006. doi: 10.1145/1180995.1181017. URL <https://doi.org/10.1145/1180995.1181017>. 1.1.1
- [19] Katherine Casey. Radical decentralization: Does community-driven development work? *Annual Review of Economics*, 10(1):139–163, August 2018. ISSN 1941-1391. doi: 10.1146/annurev-economics-080217-053339. URL <http://dx.doi.org/10.1146/annurev-economics-080217-053339>. 10.3

- [20] James I. Charlton. *Nothing About Us Without Us: Disability Oppression and Empowerment*. University of California Press, Berkeley, CA, March 1998. ISBN 9780520925441. doi: 10.1525/9780520925441. URL <https://doi.org/10.1525/9780520925441>. 2.3
- [21] Pramod Chundury, Biswaksen Patnaik, Yasmin Reyazuddin, Christine Tang, Jonathan Lazar, and Niklas Elmqvist. Towards understanding sensory substitution for accessible visualization: An interview study. *IEEE Transactions on Visualization and Computer Graphics*, 28(1):1084–1094, January 2022. ISSN 2160-9306. doi: 10.1109/tvcg.2021.3114829. URL <https://doi.org/10.1109/tvcg.2021.3114829>. 2.2.1
- [22] Pramod Chundury, Urja Thakkar, Yasmin Reyazuddin, J. Bern Jordan, Niklas Elmqvist, and Jonathan Lazar. Understanding the visualization and analytics needs of blind and low-vision professionals. In *The 26th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '24, page 1–5. ACM, October 2024. doi: 10.1145/3663548.3688496. URL <https://doi.org/10.1145/3663548.3688496>. 2.1.1
- [23] Clare Cullen and Oussama Metatla. Co-designing inclusive multisensory story mapping with children with mixed visual abilities. In *Proceedings of the 18th ACM International Conference on Interaction Design and Children*, IDC '19, pages 361–373. ACM, June 2019. doi: 10.1145/3311927.3323146. URL <https://doi.org/10.1145/3311927.3323146>. 2.2.1
- [24] Lilian de Greef, Dominik Moritz, and Cynthia Bennett. Interdependent variables: Remotely designing tactile graphics for an accessible workflow. In *Proceedings of the 23rd International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '21, page 1–6. ACM, October 2021. doi: 10.1145/3441852.3476468. URL <https://doi.org/10.1145/3441852.3476468>. 2.2.1
- [25] Deque. Automated Testing Identifies 57 Percent of Digital Accessibility Issues. Technical report, Deque, March 2021. Accessed: 2021-09-06. 3
- [26] Alan Dix. Designing for appropriation. In *Electronic Workshops in Computing*. BCS Learning & Development, September 2007. doi: 10.14236/ewic/hci2007.53. URL <https://doi.org/10.14236/ewic/hci2007.53>. 2.1.2
- [27] Sebastian Draxler, Gunnar Stevens, Martin Stein, Alexander Boden, and David Randall. Supporting the social context of technology appropriation: on a synthesis of sharing tools and tool knowledge. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, CHI '12, page 2835–2844. ACM, May 2012. doi: 10.1145/2207676.2208687. URL <https://doi.org/10.1145/2207676.2208687>. 2.1.2
- [28] Niklas Egger, Gottfried Zimmermann, and Christophe Strobbe. *Overlay Tools as a Support for Accessible Websites – Possibilities and Limitations*, page 6–17. Springer International Publishing, 2022. ISBN 9783031086458. doi: 10.1007/978-3-031-08645-8_2. URL http://dx.doi.org/10.1007/978-3-031-08645-8_2. 9.6
- [29] Danyang Fan, Alexa Fay Siu, Hrishikesh Rao, Gene Sung-Ho Kim, Xavier Vazquez, Lucy Greco, Sile O'Modhrain, and Sean Follmer. The accessibility of data visualiza-

- tions on the web for screen reader users: Practices and experiences during covid-19. *ACM Transactions on Accessible Computing*, 16(1):1–29, March 2023. ISSN 1936-7236. doi: 10.1145/3557899. URL <https://doi.org/10.1145/3557899>. 1.1.1
- [30] Stephen Few. Visual Business Intelligence - Data Visualization and the Blind. <https://www.perceptualedge.com/blog/?p=1756>, 2013. [Accessed 05-04-2025]. 1.1.1
- [31] Chris Fleizach and Jeffrey P. Bigham. System-class accessibility: The architectural support for making a whole system usable by people with disabilities. *Queue*, 22(5): 28–39, October 2024. ISSN 1542-7749. doi: 10.1145/3704627. URL <http://dx.doi.org/10.1145/3704627>. 2.2.2, 9.6
- [32] John H. Flowers, Dion C. Buhman, and Kimberly D. Turnage. Cross-Modal Equivalence of Visual and Auditory Scatterplots for Exploring Bivariate Data Samples. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 39(3):341–351, September 1997. doi: 10.1518/001872097778827151. Accessed: 2021-09-03. 2.2.1, 3
- [33] Steven L. Franconeri, Lace M. Padilla, Priti Shah, Jeffrey M. Zacks, and Jessica Hullman. The science of visual data communication: What works. *Psychological Science in the Public Interest*, 22(3):110–161, December 2021. ISSN 1539-6053. doi: 10.1177/15291006211051956. URL <https://doi.org/10.1177/15291006211051956>. 1.1.2
- [34] Krzysztof Z. Gajos, Daniel S. Weld, and Jacob O. Wobbrock. Automatically generating personalized user interfaces with SUPPLE. *Artificial Intelligence*, 174(12–13):910–950, August 2010. ISSN 0004-3702. doi: 10.1016/j.artint.2010.05.005. URL <https://doi.org/10.1016/j.artint.2010.05.005>. 2.4.3
- [35] Paula Gardner and Jess Rauchberg. *Feminist, Postcolonial, and Crip Approaches to Human-machine Communication Methodology*. Sage Publishing, 2023. 10.2
- [36] Kate S Glazko, Momona Yamagami, Aashaka Desai, Kelly Avery Mack, Venkatesh Potluri, Xuhai Xu, and Jennifer Mankoff. An autoethnographic case study of generative artificial intelligence’s utility for accessibility. In *The 25th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS ’23, page 1–8. ACM, October 2023. doi: 10.1145/3597638.3614548. URL <http://dx.doi.org/10.1145/3597638.3614548>. 10.2
- [37] Michele E. Gloede and Melissa K. Gregg. The fidelity of visual and auditory memory. *Psychonomic Bulletin & Review*, 26(4):1325–1332, April 2019. ISSN 1531-5320. doi: 10.3758/s13423-019-01597-7. URL <https://doi.org/10.3758/s13423-019-01597-7>. 1.1.1
- [38] A. Jonathan R. Godfrey, Paul Murrell, and Volker Sorge. An accessible interaction model for data visualisation in statistics. In *Lecture Notes in Computer Science*, page 590–597. Springer International Publishing, 2018. ISBN 9783319942773. doi: 10.1007/978-3-319-94277-3_92. URL https://doi.org/10.1007/978-3-319-94277-3_92. 2.2.1
- [39] David Graeber and Charlie Rose. Never cower to authority, 2006. URL <https://www.youtube.com/watch?v=qt5op6wft-8>. Accessed [2026]. 10.3

- [40] David Gray Widder, Sarah West, and Meredith Whittaker. Open (for business): Big tech, concentrated power, and the political economy of open ai. *SSRN Electronic Journal*, 2023. ISSN 1556-5068. doi: 10.2139/ssrn.4543807. URL <https://doi.org/10.2139/ssrn.4543807>. 1.1.2
- [41] Catherine Grevet and Eric Gilbert. Piggyback prototyping: Using existing, large-scale social computing systems to prototype new ones. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, CHI '15, page 4047–4056. ACM, April 2015. doi: 10.1145/2702123.2702395. URL <https://doi.org/10.1145/2702123.2702395>. 2.1.2
- [42] Karl Groves. Overlay fact sheet: What it is and how to make it work for you - karl groves, March 2023. URL <https://karlgroves.com/overlay-fact-sheet-what-it-is-and-how-to-make-it-work-for-you/>. 9.6
- [43] Nathan Hahn, Joseph Chang, Ji Eun Kim, and Aniket Kittur. The knowledge accelerator: Big picture thinking in small pieces. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, CHI'16, page 2258–2270. ACM, May 2016. doi: 10.1145/2858036.2858364. URL <https://doi.org/10.1145/2858036.2858364>. 2.1.1
- [44] Jen Hale. Extensive digitization of tactile map collection. <https://www.perkins.org/extensive-digitization-of-tactile-map-collection/>, 2016. Perkins Archives Blog, Perkins School for the Blind. July 2016. Accessed: 2022-02-23. 2.2.1
- [45] Foad Hamidi, Patrick Mbullo, Deurence Onyango, Michaela Hynie, Susan McGrath, and Melanie Baljko. Participatory design of diy digital assistive technology in western kenya. In *Proceedings of the Second African Conference for Human Computer Interaction: Thriving Communities*, AfriCHI '18, page 1–11. ACM, December 2018. doi: 10.1145/3283458.3283478. URL <https://doi.org/10.1145/3283458.3283478>. 2.1.1
- [46] A. Hamraie. Making access critical: Disability, race, and gender in environmental design. <https://belonging.berkeley.edu/aimi-hamraie-making-access-critical-disability-race-and-gender-environmental-design/>, 2019. [Online]. 2.3, 3
- [47] Aimi Hamraie. *Building Access: Universal Design and the Politics of Disability*. University of Minnesota Press, Minneapolis, MN, 2017. ISBN 9781517901639. ISBN: 978-1-5179-0163-9. 2.3, 2.4.3
- [48] Aimi Hamraie. Designing collective access: A feminist disability theory of universal design. In *Disability, space, architecture*, pages 78–297. Routledge, 2017. 2.3
- [49] Aimi Hamraie and Kelly Fritsch. Crip technoscience manifesto. *Catalyst: Feminism, Theory, Technoscience*, 5(1):1–33, April 2019. ISSN 2380-3312. doi: 10.28968/cftt.v5i1.29607. URL <https://doi.org/10.28968/cftt.v5i1.29607>. 2.3
- [50] Parker Hartman and Tim Gorichanaz. Evaluating ai-powered website accessibility overlays. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '25, page 1–4. ACM, October 2025. doi: 10.1145/366354

7.3759761. URL <http://dx.doi.org/10.1145/3663547.3759761>. 9.6

- [51] Gillian R. Hayes. The relationship of action research to human-computer interaction. *ACM Transactions on Computer-Human Interaction*, 18(3):1–20, July 2011. ISSN 1073-0516. doi: 10.1145/1993060.1993065. URL <https://doi.org/10.1145/1993060.1993065>. 10.2
- [52] Jeffrey Heer and Dominik Moritz. Mosaic: An architecture for scalable & interoperable data views. *IEEE Transactions on Visualization and Computer Graphics*, page 1–11, 2023. ISSN 2160-9306. doi: 10.1109/tvcg.2023.3327189. URL <https://doi.org/10.1109/tvcg.2023.3327189>. 1.1.2, 2.2.3, 2.4.2, 3
- [53] Ana O Henriques, Anna R. L. Carter, Beatriz Severes, Reem Talhouk, Angelika Strohmayer, Ana Cristina Pires, Colin M. Gray, Kyle Montague, and Hugo Nicolau. A feminist care ethics toolkit for community-based design: Bridging theory and practice. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, CHI '25, page 1–26. ACM, April 2025. doi: 10.1145/3706598.3713950. URL <http://dx.doi.org/10.1145/3706598.3713950>. 10.3
- [54] Thomas Hermann, Andy Hunt, and John G Neuhoff, editors. *The Sonification Handbook*. Logos Verlag Berlin, Berlin, Germany, December 2011. ISBN 9783832528195. ISBN: 978-3-8325-2819-5. 2.2.1
- [55] Jaylin Herskovitz, Andi Xu, Rahaf Alharbi, and Anhong Guo. Hacking, switching, combining: Understanding and supporting diy assistive technology design by blind people. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI '23, page 1–17. ACM, April 2023. doi: 10.1145/3544548.3581249. URL <https://doi.org/10.1145/3544548.3581249>. 2.1.2
- [56] Highsoft. Highcharts - interactive charting library for developers. <https://www.highcharts.com/>, 2009. [Online]. 2.2.1
- [57] Megan Hofmann, Devva Kasnitz, Jennifer Mankoff, and Cynthia L Bennett. Living disability theory: Reflections on access, research, and design. In *Proceedings of the 22nd International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '20, page 1–13. ACM, October 2020. doi: 10.1145/3373625.3416996. URL <http://dx.doi.org/10.1145/3373625.3416996>. 2.3, 10.2
- [58] Fred Hohman, Mary Beth Kery, Donghao Ren, and Dominik Moritz. Model compression in practice: Lessons learned from practitioners creating on-device machine learning experiences. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, CHI '24, page 1–18. ACM, May 2024. doi: 10.1145/3613904.3642109. URL <http://dx.doi.org/10.1145/3613904.3642109>. 2.1.2
- [59] Jonathan Hook, Sanne Verbaan, Abigail Durrant, Patrick Olivier, and Peter Wright. A study of the challenges related to diy assistive technology in the context of children with disabilities. In *Proceedings of the 2014 conference on Designing interactive systems*, DIS '14, pages 597–606. ACM, June 2014. doi: 10.1145/2598510.2598530. URL <https://doi.org/10.1145/2598510.2598530>. 2.1.1
- [60] Stacy Hsueh, Beatrice Vincenzi, Akshata Murdeshwar, and Marianela Ciolfi Felice. Crip-

- ping data visualizations: Crip technoscience as a critical lens for designing digital access. In *The 25th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '23, page 1–16. ACM, October 2023. doi: 10.1145/3597638.3608427. URL <http://dx.doi.org/10.1145/3597638.3608427>. 1.1.1, 2.3, 3, 10.2
- [61] Amy Hurst and Shaun Kane. Making "making" accessible. In *Proceedings of the 12th International Conference on Interaction Design and Children*, IDC '13, pages 635–638, New York, NY, USA, June 2013. Association for Computing Machinery, ACM. doi: 10.1145/2485760.2485883. Accessed: 2021-09-03. 2.1.1, 10.2
- [62] Amy Hurst and Jasmine Tobias. Empowering individuals with do-it-yourself assistive technology. In *The proceedings of the 13th international ACM SIGACCESS conference on Computers and accessibility*, ASSETS '11, page 11–18. ACM, October 2011. doi: 10.1145/2049536.2049541. URL <https://doi.org/10.1145/2049536.2049541>. 2.1.1
- [63] Liz Jackson, Alex Haagaard, and Rua Williams. Disability Dongle. <https://blog.castac.org/2022/04/disability-dongle/>, 2022. 10.2
- [64] Farnaz Jahanbakhsh, Amy X. Zhang, Karrie Karahalios, and David R. Karger. Our browser extension lets readers change the headlines on news articles, and you won't believe what they did! *Proceedings of the ACM on Human-Computer Interaction*, 6 (CSCW2):1–33, November 2022. ISSN 2573-0142. doi: 10.1145/3555643. URL <http://dx.doi.org/10.1145/3555643>. 10.3
- [65] Chutian Jiang, Wentao Lei, Emily Kuang, Teng Han, and Mingming Fan. Understanding strategies and challenges of conducting daily data analysis (dda) among blind and low-vision people. In *The 25th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '23, page 1–15. ACM, October 2023. doi: 10.1145/3597638.3608423. URL <https://doi.org/10.1145/3597638.3608423>. 2.1.1
- [66] Shuli Jones, Isabella Pedraza Pineros, Daniel Hajas, Jonathan Zong, and Arvind Satyanarayan. "customization is key": Reconfigurable textual tokens for accessible data visualizations. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, CHI '24, page 1–14. ACM, May 2024. doi: 10.1145/3613904.3641970. URL <https://doi.org/10.1145/3613904.3641970>. 2.4.3, 3
- [67] Shakila Cherise S Joyner, Amalia Riegelhuth, Kathleen Garrity, Yea-Seul Kim, and Nam Wook Kim. Visualization accessibility in the wild: Challenges faced by visualization designers. In *CHI Conference on Human Factors in Computing Systems*, CHI '22, page 1–19. ACM, April 2022. doi: 10.1145/3491102.3517630. URL <https://doi.org/10.1145/3491102.3517630>. 2.2.3, 3, 9.1, 10.2
- [68] Crescentia Jung, Shubham Mehta, Atharva Kulkarni, Yuhang Zhao, and Yea-Seul Kim. Communicating visualizations without visuals: Investigation of visualization alternative text for people with visual impairments. *IEEE Transactions on Visualization and Computer Graphics*, 28(1):1095–1105, January 2022. ISSN 2160-9306. doi: 10.1109/tvcg.2021.3114846. URL <https://doi.org/10.1109/tvcg.2021.3114846>. 2.2.1, 2.4

- [69] Shaun K. Kane, Jeffrey P. Bigham, and Jacob O. Wobbrock. Slide rule: Making mobile touch screens accessible to blind people using multi-touch interaction techniques. In *Proceedings of the 10th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS08, page 73–80. ACM, October 2008. doi: 10.1145/1414471.1414487. URL <https://doi.org/10.1145/1414471.1414487>. 2.4.2
- [70] Hyeonsu B. Kang, Xin Qian, Tom Hope, Dafna Shahaf, Joel Chan, and Aniket Kittur. Augmenting scientific creativity with an analogical search engine. *ACM Transactions on Computer-Human Interaction*, 29(6):1–36, November 2022. ISSN 1557-7325. doi: 10.1145/3530013. URL <https://doi.org/10.1145/3530013>. 2.1.1
- [71] Os Keyes, Josephine Hoy, and Margaret Drouhard. Human-computer insurrection: Notes on an anarchist hci. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, CHI ’19, page 1–13. ACM, May 2019. doi: 10.1145/3290605.3300569. URL <http://dx.doi.org/10.1145/3290605.3300569>. 10.3
- [72] Jiho Kim, Arjun Srinivasan, Nam Wook Kim, and Yea-Seul Kim. Exploring chart question answering for blind and low vision users. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI ’23, page 1–15. ACM, April 2023. doi: 10.1145/3544548.3581532. URL <https://doi.org/10.1145/3544548.3581532>. 2.2.1
- [73] N. W. Kim, S. C. Joyner, A. Riegelhuth, and Y. Kim. Accessible visualization: Design space, opportunities, and challenges. *Computer Graphics Forum*, 40(3):173–188, June 2021. doi: 10.1111/cgf.14298. URL <https://doi.org/10.1111/cgf.14298>. 2.2.3
- [74] N. W. Kim, G. Ataguba, S. C. Joyner, Chuangdian Zhao, and Hyejin Im. Beyond alternative text and tables: Comparative analysis of visualization tools and accessibility methods. *Computer Graphics Forum*, 42(3):323–335, June 2023. ISSN 1467-8659. doi: 10.1111/cgf.14833. URL <https://doi.org/10.1111/cgf.14833>. 2.4.3
- [75] David Ledo, Steven Houben, Jo Vermeulen, Nicolai Marquardt, Lora Oehlberg, and Saul Greenberg. Evaluation strategies for HCI toolkit research. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, CHI ’18, page 1–17. ACM, April 2018. doi: 10.1145/3173574.3173610. URL <https://doi.org/10.1145/3173574.3173610>. 2.1.2, 2.2.3
- [76] Zhicheng Liu and Jeffrey Heer. The effects of interactive latency on exploratory visual analysis. *IEEE Transactions on Visualization and Computer Graphics*, 20(12):2122–2131, December 2014. ISSN 1077-2626. doi: 10.1109/tvcg.2014.2346452. URL <https://doi.org/10.1109/tvcg.2014.2346452>. 2.4.2, 3, 9.6
- [77] Alan Lundgard and Arvind Satyanarayan. Accessible visualization via natural language descriptions: A four-level model of semantic content. *IEEE Transactions on Visualization and Computer Graphics*, 28(1):1073–1083, January 2022. ISSN 2160-9306. doi: 10.1109/tvcg.2021.3114770. URL <https://doi.org/10.1109/tvcg.2021.3114770>. 2.2.1, 2.4
- [78] Alan Lundgard, Crystal Lee, and Arvind Satyanarayan. Sociotechnical considerations

- for accessible visualization design. In *2019 IEEE Visualization Conference (VIS)*, page 16–20. IEEE, October 2019. doi: 10.1109/visual.2019.8933762. URL <https://doi.org/10.1109/visual.2019.8933762>. 10.2
- [79] Kelly Mack, Emma McDonnell, Dhruv Jain, Lucy Lu Wang, Jon E. Froehlich, and Leah Findlater. What do we mean by “accessibility research”? A literature survey of accessibility papers in chi and assets from 1994 to 2019. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, CHI ’21, page 1–18, New York, NY, USA, May 2021. ACM. ISBN 978-1-4503-8096-6. doi: 10.1145/3411764.3445412. URL <https://doi.org/10.1145/3411764.3445412>. Accessed: 2022-02-22. 2.2.3
- [80] Els Maeckelberghe. Feminist ethic of care: A third alternative approach. *Health Care Analysis*, 12(4):317–327, December 2004. ISSN 1573-3394. doi: 10.1007/s10728-004-6639-6. URL <http://dx.doi.org/10.1007/s10728-004-6639-6>. 10.3
- [81] Tlameo Makati, Garreth W. Tigwell, and Kristen Shinohara. The promise and pitfalls of web accessibility overlays for blind and low vision users. In *The 26th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS ’24, page 1–12. ACM, October 2024. doi: 10.1145/3663548.3675650. URL <http://dx.doi.org/10.1145/3663548.3675650>. 9.6
- [82] R G Maling and D C Clarkson. Electronic controls for the tetraplegic (possum) (patient operated selector mechanisms—p.o. s.m.). *Spinal Cord*, 1(3):161–174, November 1963. ISSN 1476-5624. doi: 10.1038/sc.1963.15. URL <http://dx.doi.org/10.1038/sc.1963.15>. 3
- [83] Jennifer Mankoff, Gillian R. Hayes, and Devva Kasnitz. Disability studies as a source of critical inquiry for the field of assistive technology. In *Proceedings of the 12th international ACM SIGACCESS conference on Computers and accessibility*, ASSETS ’10, page 3–10, New York, NY, USA, October 2010. Association for Computing Machinery, ACM. doi: 10.1145/1878803.1878807. URL <https://doi.org/10.1145/1878803.1878807>. Accessed: 2021-09-06. 10.2
- [84] Douglass L. Mansur, Merrin M. Blattner, and Kenneth I. Joy. Sound graphs: A numerical data analysis method for the blind. *Journal of Medical Systems*, 9(3):163–174, June 1985. ISSN 1573-689X. doi: 10.1007/bf00996201. URL <https://doi.org/10.1007/bf00996201>. 2.2.1
- [85] Deborah Marks. Models of disability. *Disability and Rehabilitation*, 19(3):85–91, January 1997. ISSN 1464-5165. doi: 10.3109/09638289709166831. URL <https://doi.org/10.3109/09638289709166831>. 1.1.1, 9.1
- [86] Kim Marriott, Bongshin Lee, Matthew Butler, Ed Cutrell, Kirsten Ellis, Cagatay Goncu, Marti Hearst, Kathleen McCoy, and Danielle Albers Szafir. Inclusive data visualization for people with disabilities: a call to action. *Interactions*, 28(3):47–51, April 2021. ISSN 1558-3449. doi: 10.1145/3457875. URL <https://doi.org/10.1145/3457875>. 2.2.3
- [87] David K. McGookin and Stephen A. Brewster. Soundbar: Exploiting multiple views in

multimodal graph browsing. In *Proceedings of the 4th Nordic conference on Human-computer interaction: Changing roles*, NORDICHI06, page 145–154, New York, NY, USA, October 2006. Association for Computing Machinery, ACM. doi: 10.1145/1182475.1182491. URL <https://doi.org/10.1145/1182475.1182491>. Accessed: 2021-09-03. 2.2.1

- [88] Andrew M. McNutt. No grammar to rule them all: A survey of json-style dsls for visualization. *IEEE Transactions on Visualization and Computer Graphics*, page 1–11, 2022. ISSN 2160-9306. doi: 10.1109/tvcg.2022.3209460. URL <https://doi.org/10.1109/tvcg.2022.3209460>. 2.1.2
- [89] Andrew Michael McNutt. *Understanding and Enhancing JSON-based DSL Interfaces for Visualization*. PhD thesis, University of Chicago, 2023. URL <https://doi.org/10.31219/osf.io/fy246>. 2.1.2
- [90] Catherine Mei*, Josh Pollock*, Daniel Hajas, Jonathan Zong, and Arvind Satyanarayan. Benthic: Perceptually congruent structures for accessible charts and diagrams. In *Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '25, page 1–17. ACM, October 2025. doi: 10.1145/3663547.3746342. URL <https://doi.org/10.1145/3663547.3746342>. 2.2.1
- [91] Mia Mingus. Access intimacy: The missing link. <https://leavingevidence.wordpress.com/2011/05/05/access-intimacy-the-missing-link/>, May 2011. Blog post, *Leaving Evidence*. 1.1.1, 2.3
- [92] Ehsan Mohammadi, Mike Thelwall, Stefanie Haustein, and Vincent Larivière. Who reads research articles? an altmetrics analysis of *scopus* user categories. *Journal of the Association for Information Science and Technology*, 66(9):1832–1846, April 2015. ISSN 2330-1643. doi: 10.1002/asi.23286. URL <http://dx.doi.org/10.1002/asi.23286>. 10.2
- [93] Giles Mohan and Kristian Stokke. Participatory development and empowerment: The dangers of localism. *Third World Quarterly*, 21(2):247–268, April 2000. ISSN 1360-2241. doi: 10.1080/01436590050004346. URL <http://dx.doi.org/10.1080/01436590050004346>. 10.3
- [94] Dominik Moritz, Bill Howe, and Jeffrey Heer. Falcon: Balancing interactive latency and resolution sensitivity for scalable linked visualizations. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, CHI '19, page 1–11. ACM, May 2019. doi: 10.1145/3290605.3300924. URL <https://doi.org/10.1145/3290605.3300924>. 3
- [95] Catherine A. Okoro, NaTasha D. Hollis, Alissa C. Cyrus, and Shannon Griffin-Blake. Prevalence of disabilities and health care access by disability status and type among adults — united states, 2016. *MMWR. Morbidity and Mortality Weekly Report*, 67(32):882–887, August 2018. ISSN 1545-861X. doi: 10.15585/mmwr.mm6732a3. URL <https://doi.org/10.15585/mmwr.mm6732a3>. 1.1.1
- [96] Amandalynne Paullada, Inioluwa Deborah Raji, Emily M. Bender, Emily Denton, and Alex Hanna. Data and its (dis)contents: A survey of dataset development and use in

machine learning research. *Patterns*, 2(11):100336, November 2021. ISSN 2666-3899. doi: 10.1016/j.patter.2021.100336. URL <http://dx.doi.org/10.1016/j.patter.2021.100336>. 9.6

- [97] Christopher Power, André Freire, Helen Petrie, and David Swallow. Guidelines are only half of the story: Accessibility problems encountered by blind users on the web. In *Proceedings of the SIGCHI conference on human factors in computing systems*, CHI '12, page 433–442, New York, NY, USA, May 2012. Association for Computing Machinery, ACM. doi: 10.1145/2207676.2207736. URL <https://doi.org/10.1145/2207676.2207736>. 2.2.2, 2.3, 3
- [98] Blake Ellis Reid. The curb-cut effect and the perils of accessibility without disability. *SSRN Electronic Journal*, 2022. ISSN 1556-5068. doi: 10.2139/ssrn.4262991. URL <https://doi.org/10.2139/ssrn.4262991>. 1.1.1
- [99] Rod D Roscoe, Erin K Chiou, and Abigail R Wooldridge, editors. *Advancing diversity, inclusion, and social justice through human systems engineering*. CRC Press, London, England, December 2019. 1.1.1
- [100] Heather Ruland Staines. Digital open annotation with hypothesis: Supplying the missing capability of the web. *Journal of Scholarly Publishing*, 49(3):345–365, April 2018. ISSN 1710-1166. doi: 10.3138/jsp.49.3.04. URL <http://dx.doi.org/10.3138/jsp.49.3.04>. 10.3
- [101] Manaswi Saha, Michael Saugstad, Hanuma Teja Maddali, Aileen Zeng, Ryan Holland, Steven Bower, Aditya Dash, Sage Chen, Anthony Li, Kotaro Hara, and Jon Froehlich. Project sidewalk: A web-based crowdsourcing tool for collecting sidewalk accessibility data at scale. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, CHI '19, page 1–14. ACM, May 2019. doi: 10.1145/3290605.3300292. URL <http://dx.doi.org/10.1145/3290605.3300292>. 10.3
- [102] Antti Salovaara. Inventing new uses for tools: A cognitive foundation for studies on appropriation. *Human Technology: An Interdisciplinary Journal on Humans in ICT Environments*, 4(2):209–228, November 2008. ISSN 1795-6889. doi: 10.17011/ht/urn.200811065856. URL <https://doi.org/10.17011/ht/urn.200811065856>. 2.1.2
- [103] Elizabeth B.-N. Sanders and Pieter Jan Stappers. Probes, toolkits and prototypes: three approaches to making in codesigning. *CoDesign*, 10(1):5–14, January 2014. ISSN 1745-3755. doi: 10.1080/15710882.2014.888183. URL <https://doi.org/10.1080/15710882.2014.888183>. 2.1.1
- [104] SAS. Sas Graphics Accelerator. <https://support.sas.com/software/products/graphics-accelerator/>, 2022. Accessed: 2021-09-08. 2.2.1
- [105] Arvind Satyanarayan, Dominik Moritz, Kanit Wongsuphasawat, and Jeffrey Heer. Vega-lite: A grammar of interactive graphics. *IEEE Transactions on Visualization and Computer Graphics*, 23(1):341–350, January 2017. ISSN 1077-2626. doi: 10.1109/tvcg.2016.2599030. URL <https://doi.org/10.1109/tvcg.2016.2599030>. 1.1.2, 2.2.3
- [106] Anastasia Schaadhardt, Alexis Hiniker, and Jacob O. Wobbrock. Understanding Blind

Screen-Reader Users’ Experiences of Digital Artboards. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, CHI ’21, New York, NY, USA, May 2021. Association for Computing Machinery. Accessed: 2021-09-06. 2.2.1

- [107] Doug Schepers. On underlays. Internet Archive, 2026. URL https://web.archive.org/web/20260616140431/https://www.linkedin.com/posts/daschepers_accessibility-overlays-are-good-for-business-activity-7317652815399587840-lwVL?utm_source=share&utm_medium=member_desktop&rcm=ACoAADDawBkB0doW1lI9B5DHy57VfR5jIs33Kq0. Archived version available at https://web.archive.org/web/20260616140431/https://www.linkedin.com/posts/daschepers_accessibility-overlays-are-good-for-business-activity-7317652815399587840-lwVL?utm_source=share&utm_medium=member_desktop&rcm=ACoAADDawBkB0doW1lI9B5DHy57VfR5jIs33Kq0; original URL: [linkedin.com/posts/daschepers_accessibility-overlays-are-good-for-business-activity-7317652815399587840-lwVL](https://www.linkedin.com/posts/daschepers_accessibility-overlays-are-good-for-business-activity-7317652815399587840-lwVL). Accessed: 2026-06-16. 9.6
- [108] William Schiff and Emerson Foulke, editors. *Tactual Perception*. Cambridge University Press, Cambridge, England, March 1982. ISBN 0521240956. 2.2.1
- [109] JooYoung Seo, Yilin Xia, Bongshin Lee, Sean Mccurry, and Yu Jun Yam. Maidr: Making statistical visualizations accessible with multimodal data representation. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, CHI ’24, page 1–22. ACM, May 2024. doi: 10.1145/3613904.3642730. URL <https://doi.org/10.1145/3613904.3642730>. 10.2
- [110] Hanieh Shakeri, Ye Yuan, Benett Axtell, Denise Y. Geiskkovitch, and Carman Neustaedter. Designing smart home technology for passive co-presence over distance. In *Designing Interactive Systems Conference*, DIS ’24, pages 3389–3406. ACM, July 2024. doi: 10.1145/3643834.3661508. URL <http://dx.doi.org/10.1145/3643834.3661508>. 10.3
- [111] Ather Sharif, Sanjana Shivani Chintalapati, Jacob O. Wobbrock, and Katharina Reinecke. Understanding screen-reader users’ experiences with online data visualizations. In *Proceedings of the 23rd International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS ’21, page 1–16. ACM, October 2021. doi: 10.1145/3441852.3471202. URL <https://doi.org/10.1145/3441852.3471202>. 2.2.1, 9.1
- [112] Ather Sharif, Olivia H. Wang, Alida T. Muongchan, Katharina Reinecke, and Jacob O. Wobbrock. Voxlens: Making online data visualizations accessible with an interactive javascript plug-in. In *CHI Conference on Human Factors in Computing Systems*, CHI ’22, page 1–19. ACM, April 2022. doi: 10.1145/3491102.3517431. URL <https://doi.org/10.1145/3491102.3517431>. 2.2.1
- [113] Ather Sharif, Joo Gyeong Kim, Jessie Zijia Xu, and Jacob O. Wobbrock. Understanding and reducing the challenges faced by creators of accessible online data visualizations. In *The 26th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS ’24, page 1–20. ACM, October 2024. doi: 10.1145/3663548.3675625. URL

<https://doi.org/10.1145/3663548.3675625>. 2.2.3, 3, 9.1, 10.2

- [114] Ashley Shew. *Against Technoableism: Rethinking Who Needs Improvement*. W. W. Norton, New York, NY, 2023. ISBN 9781324036661. ISBN: 978-1-324-03666-1. 2.3
- [115] Volker Sorge. Polyfilling accessible chemistry diagrams. In *Lecture Notes in Computer Science*, page 43–50. Springer International Publishing, 2016. ISBN 9783319412641. doi: 10.1007/978-3-319-41264-1_6. URL https://doi.org/10.1007/978-3-319-41264-1_6. 2.2.1
- [116] Katta Spiel, Kathrin Gerling, Cynthia L. Bennett, Emeline Brulé, Rua M. Williams, Jennifer Rode, and Jennifer Mankoff. Nothing about us without us: Investigating the role of critical disability studies in hci. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*, CHI ’20, page 1–8. ACM, April 2020. doi: 10.1145/3334480.3375150. URL <https://doi.org/10.1145/3334480.3375150>. 1.1.1, 2.3, 10.2
- [117] Clay Spinuzzi. The methodology of participatory design. *Technical communication*, 52(2):163–174, 2005. 2.1.1
- [118] Sebastian Paul Suggate. Beyond self-report: Measuring visual, auditory, and tactile mental imagery using a mental comparison task. *Behavior Research Methods*, 56(8):8658–8676, September 2024. ISSN 1554-3528. doi: 10.3758/s13428-024-02496-z. URL <https://doi.org/10.3758/s13428-024-02496-z>. 1.1.1
- [119] Cella M Sum, Rahaf Alharbi, Franchesca Spektor, Cynthia L Bennett, Christina N Harrington, Katta Spiel, and Rua Mae Williams. Dreaming disability justice in hci. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*, CHI ’22, page 1–5. ACM, April 2022. doi: 10.1145/3491101.3503731. URL <http://dx.doi.org/10.1145/3491101.3503731>. 2.3, 10.2
- [120] Cella M. Sum, Franchesca Spektor, Rahaf Alharbi, Leya Breanna Baltaxe-Admony, Erika Devine, Hazel Anneke Dixon, Jared Duval, Tessa Eagle, Frank Elavsky, Kim Fernandes, Leandro S. Guedes, Serena Hillman, Vaishnav Kameswaran, Lynn Kirabo, Tamanna Motahar, Kathryn E. Ringland, Anastasia Schaadhardt, Laura Scheepmaker, and Alicia Williamson. Challenging ableism: A critical turn toward disability justice in hci. *XRDS: Crossroads, The ACM Magazine for Students*, 30(4):50–55, June 2024. ISSN 1528-4980. doi: 10.1145/3665602. URL <http://dx.doi.org/10.1145/3665602>. 2.3, 10.2
- [121] Sarit Felicia Anais Szpiro, Shafeka Hashash, Yuhang Zhao, and Shiri Azenkot. How people with low vision access computing devices: Understanding challenges and opportunities. In *Proceedings of the 18th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS ’16, page 171–180, New York, NY, USA, October 2016. ACM. ISBN 9781450341240. doi: 10.1145/2982142.2982168. URL <https://doi.org/10.1145/2982142.2982168>. 2.2.3
- [122] Pierre Tchounikine. Designing for appropriation: A theoretical account. *Human-Computer Interaction*, 32(4):155–195, July 2016. ISSN 1532-7051. doi: 10.1080/07370024.2016.1203263. URL <https://doi.org/10.1080/07370024.2016.1203263>. 2.1.2

- [123] Yuanyang (YY) Teng and Darren Gergle. Access is not enough: Toward developmental flourishing. In *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*, CHI EA '26, page 1–6. ACM, April 2026. doi: 10.1145/3772363.3798552. URL <http://dx.doi.org/10.1145/3772363.3798552>. 1.1.1
- [124] John R Thompson, Jesse J Martinez, Alper Sarikaya, Edward Cutrell, and Bongshin Lee. Chart reader: Accessible visualization experiences designed with screen reader users. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI '23, page 1–18. ACM, April 2023. doi: 10.1145/3544548.3581186. URL <https://doi.org/10.1145/3544548.3581186>. 2.2.1, 2.4.1
- [125] Alexandra To, Angela D. R. Smith, Dilruba Showkat, Adinawa Adjagbodjou, and Christina Harrington. Flourishing in the everyday: Moving beyond damage-centered design in hci for bipoc communities. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*, DIS '23, pages 917–933. ACM, July 2023. doi: 10.1145/3563657.3596057. URL <https://doi.org/10.1145/3563657.3596057>. 2.1.1, 10.2
- [126] Visa. Visa Chart Components. <https://github.com/visa/visa-chart-components>, 2022. Accessed: 2022-12-01. 2.2.1
- [127] W3C. Web content accessibility guidelines (wcag) 2.1. <https://www.w3.org/TR/WCAG21/>, 2018. Accessed: 2021-09-03. 2.2.2, 3
- [128] WAI. Understanding success criterion 4.1.2: name, role, value. WCAG standard, W3C, 2016. Accessed: 2022-03-04. 2.2.2
- [129] Bruce N. Walker and Michael A. Nees. Theory of sonification. In Thomas Hermann, Andy Hunt, and John G. Neuhoff, editors, *The Sonification Handbook*, chapter 2, pages 9–39. Logos Verlag, Berlin, Germany, 2011. 2.2.1
- [130] Chris Weaver. Multidimensional visual analysis using cross-filtered views. In *2008 IEEE Symposium on Visual Analytics Science and Technology*, page 163–170. IEEE, October 2008. doi: 10.1109/vast.2008.4677370. URL <https://doi.org/10.1109/vast.2008.4677370>. 3
- [131] WebAIM. Webaim: The WebAIM Million - An annual accessibility analysis of the top 1,000,000 home pages. <https://webaim.org/projects/million/#wcag>, 2021. Accessed: 2021-09-03. 10.3
- [132] David Gray Widder and Richmond Wong. Thinking upstream: Ethics and policy opportunities in ai supply chains, 2023. URL <https://doi.org/10.48550/arxiv.2303.07529>. 1.1.2
- [133] David Gray Widder, Meredith Whittaker, and Sarah Myers West. Why ‘open’ ai systems are actually closed, and why this matters. *Nature*, 635(8040):827–833, November 2024. ISSN 1476-4687. doi: 10.1038/s41586-024-08141-1. URL <https://doi.org/10.1038/s41586-024-08141-1>. 1.1.2
- [134] Rua M. Williams and LouAnne E. Boyd. Prefigurative politics and passionate witnessing. In *Proceedings of the 21st International ACM SIGACCESS Conference on Computers and*

- Accessibility*, ASSETS '19, page 262–266. ACM, October 2019. doi: 10.1145/3308561.3355617. URL <http://dx.doi.org/10.1145/3308561.3355617>. 10.2
- [135] Rua M. Williams, Kathryn Ringland, Amelia Gibson, Mahender Mandala, Arne Maibaum, and Tiago Guerreiro. Articulations toward a crip hci. *Interactions*, 28(3):28–37, April 2021. ISSN 1558-3449. doi: 10.1145/3458453. URL <http://dx.doi.org/10.1145/3458453>. 2.3, 10.2
- [136] Rua Mae Williams, Louanne Boyd, and Juan E. Gilbert. Counterventions: a reparative reflection on interventionist hci. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI '23, page 1–11. ACM, April 2023. doi: 10.1145/3544548.3581480. URL <http://dx.doi.org/10.1145/3544548.3581480>. 10.2
- [137] B. L. Wimer, L. South, K. Wu, D. A. Szafir, M. A. Borkin, and R. A. Metoyer. Beyond vision impairments: Redefining the scope of accessible data representations. *IEEE Transactions on Visualization and Computer Graphics*, 30(12):7619–7636, December 2024. 2.2.3
- [138] Langdon Winner. Do artifacts have politics? In *Computer Ethics*, page 177–192. Routledge, May 2017. ISBN 9781315259697. doi: 10.4324/9781315259697-21. URL <https://doi.org/10.4324/9781315259697-21>. 1.1.2
- [139] Jacob O. Wobbrock, Shaun K. Kane, Krzysztof Z. Gajos, Susumu Harada, and Jon Froehlich. Ability-based design: Concept, principles and examples. *ACM Transactions on Accessible Computing*, 3(3):1–27, April 2011. ISSN 1936-7236. doi: 10.1145/1952383.1952384. URL <https://doi.org/10.1145/1952383.1952384>. 1.1.2, 2.1.2, 2.4.3
- [140] Mintesnot Woldeamanuel and Andrew Kent. Measuring walk access to transit in terms of sidewalk availability, quality, and connectivity. *Journal of Urban Planning and Development*, 142(2), June 2016. ISSN 1943-5444. doi: 10.1061/(asce)up.1943-5444.0000296. URL [http://dx.doi.org/10.1061/\(ASCE\)UP.1943-5444.0000296](http://dx.doi.org/10.1061/(ASCE)UP.1943-5444.0000296). 10.3
- [141] R. Wood et al. “creatures of habit”: influential factors to the adoption of computer personalization and accessibility settings. *Universal Access in the Information Society*, 23(2): 927–953, March 2023. 2.4.3, 3
- [142] Rebecca Yeo and Karen Moore. Including disabled people in poverty reduction work: “nothing about us, without us”. *World Development*, 31(3):571–590, March 2003. ISSN 0305-750X. doi: 10.1016/s0305-750x(02)00218-8. URL [https://doi.org/10.1016/s0305-750x\(02\)00218-8](https://doi.org/10.1016/s0305-750x(02)00218-8). 1.1.1
- [143] Haixia Zhao, Catherine Plaisant, Ben Shneiderman, and Jonathan Lazar. Data sonification for users with visual impairment: A case study with georeferenced data. *ACM Transactions on Computer-Human Interaction*, 15(1):1–28, May 2008. ISSN 1557-7325. doi: 10.1145/1352782.1352786. URL <https://doi.org/10.1145/1352782.1352786>. 2.2.1
- [144] Jonathan Zong. Using real names of disabled participant-contributors to practice citational justice in accessibility. In *2025 IEEE Workshop on Accessible Data Visualization (Ac-*

cessViz), page 30–33. IEEE, November 2025. doi: 10.1109/accessviz68666.2025.00011. URL <https://doi.org/10.1109/accessviz68666.2025.00011>. 2.3

- [145] Jonathan Zong, Crystal Lee, Alan Lundgard, JiWoong Jang, Daniel Hajas, and Arvind Satyanarayan. Rich screen reader experiences for accessible data visualization. *Computer Graphics Forum*, 41(3):15–27, June 2022. doi: 10.1111/cgf.14519. URL <https://doi.org/10.1111/cgf.14519>. 2.2.1, 2.3, 2.4.1, 2.4.2, 3
- [146] Jonathan Zong, Isabella Pedraza Pineros, Mengzhu (Katie) Chen, Daniel Hajas, and Arvind Satyanarayan. Umwelt: Accessible structured editing of multi-modal data representations. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, CHI ’24, page 1–20. ACM, May 2024. doi: 10.1145/3613904.3641996. URL <https://doi.org/10.1145/3613904.3641996>. 2.2.1